

TESLA INC TSLA

April 30, 2024 - 1:25pm EST

by	Weighing Machine Capital			
			2024	2025
Price:	183.80	EPS	\$2.56	\$3.38
Shares Out. (in M):	3,484	P/E	72	54
Market Cap (in \$M):	640,400	P/FCF	0	0
Net Debt (in \$M):	-21	EBIT	0	0
TEV (in \$M):	618	TEV/EBIT	0	0

Description

Tesla is the most controversial company I've seen in 30 years in the investment business. Believers in Tesla are cult-like just as TSLAQ members adamantly believe Elon Musk belongs in prison. It's reasonable to be on either side of this debate.

Tesla will become the most valuable company in the world if it successfully solves autonomous transportation. If it does not, TSLA stock is going down 90%.

Autonomous transportation is really hard to solve. Whoever succeeds will create trillions of dollars of value.

I strongly encourage anyone who wants to debate this FIRST go to their nearest Tesla delivery center and test drive the Full Self Driving (FSD) software for themselves. I've been using FSD Beta since its release in 2020, and frankly, it kinda sucked until a few months ago. Then everything changed with a complete overhaul of the software from a heuristics/C++ code-based architecture to a full neural net design.

Tesla is the leading real-world robotics/AI company that is now on a path to solving autonomous transportation. Many think that Tesla's Full Self Driving ("FSD") software is vaporware, mainly because it's been promised to drive better than a human for many years now, and has failed to deliver that promise thus far.

That perception is about to change. Tesla began redesigning its autonomous driving software stack to an entirely "end to end" neural network architecture in the latter half of 2022. Prior to this period, it was using primarily a heuristics/human code based architecture that was suboptimal in terms of reaching human level, and eventually, beyond human level driving capability.

FSD version 12 of the software of this full neural network architecture was released to the public a few months ago and it is an absolute game changer. I have personally been using first Autopilot, then FSD Beta since 2019, and was concerned with the rate of progress due to the limitations of the C++ code architecture. That architecture is now fully neural network which is the reason I strongly believe FSD is now on the path to achieving super-human driving capability.

"If you don't disrupt yourself, someone else will do it for you."

-Salim Ismail, from his book Exponential Organizations.

I began to learn in the Fall of 2022 that they were throwing away the old code-based system in favor of full neural nets. This is when I became very bullish on Tesla. Though it was very costly to do so, and would set the progress back a few years throwing away 8 years of software code, it was a bold decision; a "bet the company" decision.

Today Tesla is considered just a car company. I strongly disagree. It has an insurmountable lead in real-world driving data and AI architecture that is training their neural network driving models. No other company has anything close to this. There are today 5 million robot cars driving around the world capturing video data and sending it back to Tesla HQ to train their neural network AI driving models.

This new age of Artificial Intelligence that was launched a year ago with the release of Open AI's Large Language Model, Chat GPT-3, will change the economy in ways we can only imagine.

Beyond-human level intelligence will be brought to every machine and software application you could think of because of artificial intelligence.

My thinking is that machine intelligence is going to be a lot more difficult to build than software intelligence. Since last year's Chat GPT moment, there have been numerous copy cats that have been released such as Google's Gemini, Meta's Llama, Microsoft's Copilot, Perplexity's Claude 3, and X.ai's Grok. They are all essentially equivalent in terms of accuracy, since, they're all trained on the same data set, namely every website that exists on the open internet. The only differences among them is their speed and that comes down to size of their neural network models and differences in compute power.

Real-world, machine AI is going to be very different in my opinion. Whereas software AI has very little barrier to entry as evidenced by the many LLM chatbots that have sprung up in just the last year; real-world AI is extremely difficult both in terms of access to data, and in terms of manufacturing the machines at scale and cost-effectively.

Chatbots were the first application of "software" AI. In my strong opinion, autonomous cars will be the first application of "real-world" AI and Tesla will be the clear winner.

No other automobile manufacturer in the world, except for China's BYD, builds electric cars at a positive profit margin besides Tesla. It takes the scale of millions of electric vehicles produced per year to build EVs profitably. It's the classic innovator's dilemma. If you're Toyota, Ford or GM, and you're losing \$50k on every electric vehicle produced due to your lack of scale, you stick to what satisfies your shareholders who demand profits; making profitable cars/trucks with combustion engines.

Electric drive-train cars are going to continue to gain market share until they're 100% of the fleet at some point in the next 10 years. Why? Not their smaller environmental footprint, though that's a positive externality. It's because of cost. Electric cars have only 10% of the parts of a combustion engine car. No oil, no transmission, no engine. Just electrons from the battery to the motor, and power out to turn the axles. Much more efficient and cost effective than combustion engines. The limiting factor so far has been the cost of the battery. But, today, that cost is down to below \$100 per kilowatt hour from \$300 just 7 years ago. The battery accounts for nearly 40% of the cost of an electric vehicle so as this cost continues to drop, in a few years, electric vehicles will be cheaper than their combustion engine equivalents.

Tesla is the leader in EV battery technology and scale. Scale is very important. The more cars and batteries you manufacture, the more you benefit from economies of scale and no one has the scale that Tesla has in EVs and batteries.

"Be fearful when others are greedy and be greedy when others are fearful"

-Warren Buffett

Just a year and a half ago, Nvidia was just a company that made chips for gaming and cryptocurrency mining. Today, investors have ultimately figured out that Nvidia is more of a software platform company that is providing the picks and shovels for the 21st century AI gold rush. Nvidia stock is up 8 fold since it bottomed near \$100 just 18 months ago. Nvidia's business was suffering at that time even though, under the surface, they had been building an AI powerhouse on top of their CUDA software optimization layer for decades. Unfortunately, I missed it, since I don't focus on chip companies. I think Tesla has a good chance to do something similar over the next few years if I'm right about their real-world AI efforts.

Yes, Tesla's business is having a tough first half of 2024. Today, Tesla appears to be just a car company similar to how a year and a half ago, Nvidia was just a gaming/crypto chip company. Musk's political tweets turn away many potential customers and there's an air pocket right now before the next generation \$25k robotaxi "for the masses" launches in 2025.

Tesla's stock is down from \$300 last summer to \$185 due to this temporary setback as the rest of the Magnificent 7 Mega-Caps have gone straight up.

I thought Q1 earnings last week were going to be a disaster. They actually were not that bad. I was modeling a loss for Q1 given the 9% YoY decline in Q1 deliveries/revenue. Tesla actually made \$.45 in Non-GAAP EPS in Q1. This is quite remarkable given the fixed cost deleveraging in a manufacturing business.

The TAM (Total Addressable Market) of real-world AI is easily in the tens of trillions of dollars. I believe we're going to have a Chat GPT moment in real-world AI before the year is out; and it will be Tesla's autonomous driving platform that it's been building for almost a decade.

Why this year? Two words, transformers and compute power.

Transformers, (the T in Chat GPT), ushered in the age of AI in 2017 when Google invented this novel approach to deep learning. ["Attention is All you Need"](#) (PDF). Advances in Neural Information Processing Systems.

Sufficient compute power to fully utilize this revolutionary approach to AI took a few more years. With Nvidia's release of the A100 Tensor Core GPU in 2020, it was now possible to train a multi-billion parameter AI model to memorize and understand all of Wikipedia and soon the entire internet with their next generation H100 GPU.

Tesla uses this same approach to deep learning utilizing transformers and immense compute power to train its autonomous driving platform.

Why can't someone else replicate this approach? They certainly can and they are. Google's Waymo, GM's Cruise, and many others are using this combination of transformers and massive compute power to train autonomous driving systems.

So why bullish on Tesla?

NO ONE HAS THE REAL WORLD DATA THAT TESLA HAS.

5 million Tesla cars driving around with 8 cameras each ingesting video data and sending back to Tesla HQ to train their neural nets.

And even if competitors had the data, which they don't, they don't manufacture the robot cars at scale or profitably. Waymo, arguably closest to Tesla in the autonomous driving race, only has approximately 1,000 cars driving around. And their cars are not everywhere in the world like Tesla's. Instead they're geofenced in easy driving environments such as Phoenix AZ, San Francisco CA, and Austin TX. And Waymo cars cost over \$100,000 to produce and Waymo doesn't manufacture them, partners do it for me.

Tesla sells the cars at a profit and every car they sell contains all the equipment necessary to train the FSD system regardless of whether a buyer subscribes to the software or not. They also get the data regardless of whether FSD is turned on or not. Today, only about 8% of the Tesla cars on the road have FSD turned on. But, they get data from 100% of the cars.

Not only does Tesla have the data advantage from 5 million cars driving around, they have another huge advantage, the human driver supervisors. Every driver of those 5 million Tesla cars, is a free software debugger for Tesla. The billions of miles of real-world driving data that is sent back to Tesla HQ is extremely important to training the AI models, but what is even more important is the data generated from when the human drivers dis-engage FSD.

The 900,000 human drivers who use FSD is the most valuable asset to Tesla. When they dis-engage FSD, they notify Tesla HQ that they disagreed with the

autonomous action the car took. This video data is flagged to Tesla HQ and the neural network polls the rest of the fleet to send back similar looking scenes to the ones that are flagged as a disengagement.

It really is amazing that Tesla charges FSD subscribers \$8,000 or \$99/mo. for the software when they actually gain a huge asset for every buyer of the software since the most important data that improves the system is the disengagement data provided by the human drivers.

This is creating an important network effect and costs less than zero for Tesla to create it. As more drivers subscribe to FSD, more disengagement data is sent to Tesla HQ to improve FSD. The better FSD gets, the more people will want to subscribe to FSD, leading to more subscribers, more disengagement data, and further improvements to the FSD system, etc., etc.

FSD is a game changer. 40,000 Americans and 2,000,000 people around the world die from car crashes each and every year.

I have a Google Pixel phone instead of an iPhone because my Android experience is not much different from an iOS experience.

It's going to be very different when it comes to choosing your autonomous driving system. No one is going to want the second best autonomous driving system. Loved ones' lives are at risk.

It's going to be a 'winner takes all' market I believe. That market is in the trillions of dollars when you think about the size of the world's road-based transportation market. Trillions.

When FSD gets to a level that is superior to human level driving, anyone who wants to get it will have to buy a Tesla car. At that moment that FSD reaches super-human capability, which I believe happens before the end of 2024, investors will wake up to what Tesla's been building.

Think the iPhone. It's not better than a Samsung or a Lenovo, etc in terms of nuts and bolts of the actual phone. It's what you can do with it. The App Store makes everyone buy an iPhone.

I think the same thing happens with Tesla. FSD is to Tesla as the App Store is to iPhone.

Everyone will want a Tesla to get access to FSD.

And that's only the first application of real-world AI. The next application will be humanoid robots. Then we're talking about tens of trillions of dollars of market value. Though it's certainly early in that race, I think Tesla has advantages there as well. They've already begun manufacturing humanoid robots and are optimizing for low cost and power efficiency. The same system architecture Tesla is using for FSD, they're using for humanoid robots. That's very important when it comes to scale. Other companies have to build a brand new system for humanoid robots whereas Tesla does not; its system for robots is already in place and at scale with regards to compute and neural network architecture optimization.

Tesla's transition of it's autonomous driving software stack to full 'end to end' neural nets, combined with 900,000 human supervisors that label edge cases by disengaging FSD, will lead to a flywheel allowing FSD to rapidly improve in the months ahead. A year from now, FSD will be 5X safer than an average human driver.

Consumers will want FSD. And the only way to get it is to buy a Tesla car....

The iPhone moment for the transportation industry.

I do not hold a position with the issuer such as employment, directorship, or consultancy.

I and/or others I advise hold a material investment in the issuer's securities.

Catalyst

FSD gets beyond human level safety within a year and millions of consumers order a Tesla car to get access to it.

Messages

Subject valuation
Entry 04/30/2024 01:56 PM
Member churchill

Thanks for the write up.

Why is \$600b an appealing valuation? What kind of profits does Tesla need to generate to make this work and how and when do we get there?

Subject Thanks!
Entry 04/30/2024 02:14 PM
Member bowd57

Hi, Weighing Machine Capital --

It's great to see the bull case put out. I might have some quibbles with your take on the auto side, but don't think them worth getting into, because I disagree with this part:

> Today, Tesla appears to be just a car company

To me, Tesla looks much more like an AI/autonomy company with a legacy auto business, I think most of the stock value can be attributed to that rather than auto.

Since this is a reactivation idea, I'm assuming you haven't been really been able to follow the discussion on my (now superceded, thank God) writeup because I try to keep my messages members-only. Really briefly:

1: I suggest using the teslafsdtracker --

<https://www.teslafsdtracker.com/>

to assess Tesla's progress. That currently shows about 160 city miles to critical disengagement for FSD 12.x, which I take to be the key metric for robotaxi/personal chauffeur purposes.

2: At 1 accident/10 critical disengagements, FSD needs to 1000x that number to equal a non-impaired human at-fault accident rate.

3: To quote myself from earlier in the day:

Me:> I think Tesla needs to improve MTBF by 1000x. I'm open to different numbers. If they can 1000x that in a year they'll have a robotaxi in a year. if they can 10x that every year, they'll have a robotaxi in 3 years. If they can 4x that every year, they'll have a robotaxi in 5 years. If they can 2x that every year, they'll have a robotaxi in 10 years. There's a separate conversation to be had about "with the current fleet".

, so I think they need to show rapid progress, progress that should be visible in the tracker, in order for the program to be commercially relevant.

There's been a lot of back and forth on this. There are some other issues to discuss. But the "other issues" are secondary to Tesla needing to improve the product. Is that a framework that makes sense to you and that you can work with? Does it make sense to you at all? Am I looking at this the wrong way?

Yours,

Bowd

Subject Data/Cost
Entry 04/30/2024 02:17 PM
Member abcd1234

You say that Tesla having people use their FSD to improve the AI is the biggest advantage. If this is such an advantage and so important, why aren't they giving it to drivers for free? I know they just offered 1 month free for everyone but why stop at 1 month? Is it the compute cost that Tesla has to incur when digesting that data which forces them to charge for it?

Subject Re: valuation
Entry 04/30/2024 02:20 PM
Member Weighing Machine Capital

Tesla is releasing a new version (training a new neural network model) every 2-3 weeks.

Tesla has already spent approximately \$40 Billion by my estimates in loading all the necessary hardware (sensors + inference computer) on > 5 million cars in the fleet. Every incremental FSD subscriber is mostly pure profit from hereon. (> 90% incremental gross margin)

You really should try it out. <https://www.tesla.com/>

Click on "demo drive"

and when you arrive for it, make sure to test drive a car that has FSD enabled. It will blow your mind.

The business model is this. In my opinion, FSD is already safer than the average human. It's also a much better experience turning on FSD and not having to drive the car manually yourself.

As the software improves (and it's iterating very fast with the architecture switch to end to end neural nets), more and more people will get the memo.

They'll buy the car to get FSD. Tesla sales will rocket higher. The profits will rocket higher given the leverage in the manufacturing of the cars and the pure profit in the FSD software.

Demo the FSD software for yourself. Demo it for 3 hours, not 30 minutes.

Subject	Re: Thanks!
Entry	04/30/2024 02:43 PM
Member	Weighing Machine Capital

Hi Bowd,

Have you tried FSD version 12.3.4?

I agree with you about 100X or 1000X the reduction of safety critical disengagements.

The architecture change to neural nets from heuristics with the launch of 12.x is the key to accelerating this progress. That comes down to four components of the architecture.

1. Increasing the amount of data used to train the model
2. Increasing the compute power to train the model
3. Increasing the size of the model (increasing the # of parameters or nodes in the model)
4. Increasing the # of hours used to train the model

There are always engineering tradeoffs with neural nets because the software is probabilistic, not rule-based. Also, simply 10x-ing the amount of data doesn't 1/10x the number of disengagements due to diminishing scaling laws with neural net training as the models approach the "perfection" asymptote. (Neural net software will never be perfect)

FSD had to go to a neural network architecture because it's impossible to hardcode for every single edge case in the real world driving environment. It will never be perfect, but it doesn't have to be. It just has to be much better than human driving. Then more people subscribe to FSD as it improves, resulting in more and more data, which helps it improve faster with more and more disengagement data.

Subject	Re: Data/Cost
Entry	04/30/2024 02:57 PM
Member	Weighing Machine Capital

You bring up a good point. Why doesn't Tesla incentivize as many FSD subscriptions as possible?

The reason is this. When FSD sucked up until the changeover from heuristics to end to end neural nets, Tesla ONLY wanted passionate FSD fanatics who really understood the software to use it. Because, it was not that good, and it would cause accidents if a driver didn't understand the software really well.

The switch to v.12 neural nets dramatically improved the safety, so now, Tesla can release it to more people, beyond the fanatics, to let's call them enthusiasts by charging \$99/month.

When it gets even safer in a few months, either it will organically get a higher take rate, or Tesla will drop the price even further to get after more data.

The thing to understand about neural network scaling is that as the performance gets closer and close to the asymptotic "perfect" line, the amount of data/compute/model size has to grow exponentially in order to get incremental improvements.

So, either, they'll get much more data because the public wants to subscribe, or they'll give it away for free. In the long term plan, it doesn't matter.

Tesla will have an 85,000 H100 GPU cluster by the end of the year - plenty of compute power.

Tesla has the very best machine learning engineers to continually optimize the training models.

Most importantly, Tesla has a fleet of now 900,000 human drivers that are labeling the edge case disengagements. This is the secret sauce that no other player has.

Subject	At what point will AD be "good enough" to get the green light?
Entry	04/30/2024 03:11 PM
Member	katana

WeighingMachine says: FSD "will never be perfect, but it doesn't have to be. It just has to be much better than human driving."

This issue is not specific to Tesla but obviously affects Tesla: How safe does AD have to get to be accepted by the regulators and the public? For a technology adoption this huge, humans *clearly* have a strong bias to focus on the downsides of the New Thing and ignore the already-existing downsides of the Old Thing - even when the downsides are on the exact same dimension. Such as accidents (or deaths) per mile (or any other unit-of-driving metric).

It seems fairly clear that, to get AD accepted, it's going to have to be proven to be, as WeightMachine says, "much better" (i.e. safer) than human driving. Is there some sort of generally-accepted standard of how safe that will be? I know from Bowd throwing around some relevant hard numbers that perhaps there is such a number already and I simply haven't been paying enough attention.

Next level of detail, and back to Tesla: Would moving that acceptable number higher or lower, within a reasonable range of what will transpire, materially change the odds of which AD provider will get there first?

Subject	Re: At what point will AD be "good enough" to get the green light?
Entry	04/30/2024 03:23 PM
Member	kgalt

katana what is AD?

Subject	Re: At what point will AD be "good enough" to get the green light?
Entry	04/30/2024 03:36 PM
Member	Weighing Machine Capital

@katana,

I agree with what you said, and you said it eloquently, *"For a technology adoption this huge, humans clearly have a strong bias to focus on the downsides of the New Thing and ignore the already-existing downsides of the Old Thing - even when the downsides are on the exact same dimension."*

I don't think it matters in the end though.

FSD v. 12.3.4 is already close to human level. But, it's more about whether it's on a path to improving at a reasonable iteration speed that will march to ultimately, 99.9999999% accuracy. That's the level that's needed for Level 5 autonomy. Tesla is already testing internally v. 12.4 and 12.5. These are not dot releases so I think they will be step-change improvements.

I think a lot will be clearer about the iteration speed and path once these versions are publicly released in the coming month. I think the public will get v. 12.4 within 2 weeks, and v. 12.5 about 3 weeks after that. So by the end of May, we'll have a pretty good sense of the trajectory of FSD.

As you say, autonomous driving technology will be one of the, if not the, biggest technology change we've ever seen.

The multi-trillion dollar questions are:

Who will solve it first? When will it be solved? What first solver advantages are there?

Tesla has this magical fly wheel of 900k actual humans disengaging (flagging) bugs in the software, a 35k H100 compute cluster training on this data, and a fleet of > 5 million cars with full sensor stack/inference computers operating in shadow mode whereby even if FSD is not subscribed, the human drivers are flagging edge case bugs when the control output disagrees with FSD shadow mode predictions.

This fly wheel is impossible to replicate at this point in my opinion by any of the other AD players.

Subject	Re: Re: At what point will AD be "good enough" to get the green light?
Entry	04/30/2024 03:39 PM
Member	Weighing Machine Capital
@jgalt	

AD = Autonomous Driving

Subject	Needs 1000x or "Already Better Than Human"?
Entry	04/30/2024 06:28 PM
Member	bowd57
Hi, WMC --	

I'm trying to figure out what you believe the current safety profile of FSD is.

To Churchill:> In my opinion, FSD is already safer than the average human.

To Katana:> FSD v. 12.3.4 is already close to human level

To Me:> I agree with you about 100X or 1000X the reduction of safety critical disengagements.

My 1000x was to get to human level. I'm guessing you think Tesla is there or close, but needs to show superhuman safety for RT purposes? I'm really not sure. If you think that FSD is human level right now, I'm not sure what to say, other than, join TMC, start a thread called "FSD Is Already At Human Safety Levels", and be prepared to be shouted down by the fans if you get any response at all.

Sorry if that sounded confrontational. The main point is, I think FSD needs 1000x to get to human levels. I'm not going to start WWII over 100x either; in my framework, that just shaves a year or two off delivery times. As long as we're agreed that it needs to get a lot better, on a short time frame, and we can use the tracker to judge progress, I'm not going to argue about the details. If we agree on the basics, I'd like to move on to speculating about the (right now!) unknowable.

Yours,

Bowd

Subject	waymo
Entry	04/30/2024 08:50 PM
Member	finn520
What does Waymo leadership/engineers say about the TSLA approach and why they don't think it will work? Or are they so path dependent from so many years pursuing their approach that you think they have their heads in the sand?	

Subject	Unprotected left turn
Entry	04/30/2024 10:56 PM
Member	yxd0950

<https://www.youtube.com/watch?v=-qBme0ddng&t=439s>

Munger said that you should pay attention especially to facts disconfirming to your thesis. This, as Bowd suggested before, is indisputable utter fail even with the latest ver. If users have to switch to manual for every single unprotected left turn, FSD's utility could be drastically reduced. Optimists will argue that future versions should be able to address this issue, which, if turns out to be true, will prove that the new AI system could potentially solve many other hard self-driving issues. Only time will tell but this left turn test should be one vital data point both bulls and bears keep their eyes on as future FSD versions release.

Subject	Cost of data?
Entry	05/01/2024 12:32 AM
Member	Nails4

Dumb question, but how does Tesla actually gather and process the data gathered by cars on the road? I have to imagine each car is generating an insane amount of sensor, video, and other telemetry data. Who's paying to have this data delivered to Tesla? The bandwidth costs alone must be significant?? How do they track and label the problems? Do drivers really do this manually?

I guess the key argument isn't how good FSD is today, but rather how quickly it improves. The difference between Bowd and the author seems to be that Bowd thinks the "good enough" timeline is many years, but the author seems to be arguing for a much shorter frame — that the end state will be clear within months... and that's when the narrative in the stock will change radically

Subject	Re: Needs 1000x or "Already Better Than Human"?
Entry	05/01/2024 09:23 AM
Member	Weighing Machine Capital

@Bowd,

Have you experienced any FSD version?

FSD Tracker, in theory, is a great way to track progress except for potential selection bias.

Subject	Re: Cost of data?
Entry	05/01/2024 10:04 AM
Member	kgalt

Nails4:

My guess it Tesla has been paying for the bandwidth. Tesla used to employ thousands of workers to manually label the video, because the video quality has to be good (meaning, you have to feed the machine GOOD drivers doing good things, not shitty drivers, otherwise FSD won't learn how to drive well), but it has since fired a bunch (all?) of those humans as the software is doing auto-labeling.

"Do drivers really do this manually?" — what do you mean?

Bowd "thinks" a lot but really should try FSD himself. There's a difference between looking at stats and MTBF and disengagements and then experiencing the actual thing.

For a lot of people and uses cases the actual thing (FSD) is already good enough and it's only going to get better.

For a discussion on rate of improvement, listen to the most recent earnings call.

Subject	TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/01/2024 10:15 AM
Member	Weighing Machine Capital

Waymo uses LIDAR / Vision / HD maps vs. Tesla using vision only.

In my opinion, Waymo will remove LIDAR soon in order to reduce the cost of the sensor suite when it decides to scale the number of vehicles. It used LIDAR to build their foundational model. Once they have solved for the software part of the equation, they will solve for the hardware part (reasonable cost in order to scale)

In my opinion, autonomous driving can be solved with vision only. The transportation infrastructure around the world is designed for humans with two cameras (eyes) capturing video input and a biological neural net (brain). 8 high resolution cameras and a synthetic neural net is enough in my opinion. It just needs a lot of edge-case data and training.

Transformers changed everything. With the advances deep learning neural nets get from using transformer architecture, it comes down to scaling edge-case data, compute, and model parameter size. This was not possible even a few years ago. Compute wasn't powerful enough, and transformer architecture didn't exist, or better said, it took time for the machine learning community to apply transformers to machine learning tasks such as generative AI/computer vision.

The debate about whether Tesla's autonomous driving architecture/strategy will succeed in being sufficiently better than human is a fascinating one. It's not about where the software is today, it's an engineering debate about whether the system is architected in such a way that will inevitably improve the software sufficiently over a reasonable amount of time.

The secret sauce is the fleet of human supervisors labeling the bugs. 100s of thousands of FSD subscribers are disengaging when FSD makes a mistake. That's the critical data Tesla needs to improve from 99% accurate to 99.9999999% accurate. Tesla has what they refer to a "fleet triggering system". For example, when they get a disengagement, they send it to a neural net (distinct from their main foundational model) that first determines the safety critical "score" of the disengagement. If it is above a threshold for safety critical, the fleet is then polled for video clips similar to the disengagement scenario in order to gather as much data as possible to re-train the foundational model to "learn" how humans handle the scenario that resulted in the disengagement (hope that made sense).

That's the path needed to solve autonomous driving in my opinion. No one else has this architecture. But, most importantly, no one else has the **fleet of 100s of thousand of human bug labelers**. And Tesla gets paid by these human labelers! How Tesla was able to design this is one for the HBS case study record books.

Subject	Re: Re: Cost of data?
Entry	05/01/2024 12:15 PM
Member	Sextant

I don't own a Tesla, so this may be a stupid question. But 1) wouldn't the customer need to sign some privacy consent? Presumably the data would also involve video that may capture the customer, surrounding, home etc.?

2) wouldn't the customer see massive upload data transfers when the car is parked at home and uploading data during idle time? Haven't seen that talked about anywhere as an issue/observation.

Subject	Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/01/2024 12:39 PM
Member	jpgalt

I disagree that Waymo is going to get rid of LIDAR. It would mean they'd have to change their entire approach.

Subject	Re: Re: Re: Cost of data?
Entry	05/01/2024 12:41 PM
Member	jpgalt

Yes, of course you sign off on privacy, it's in the car's settings. There's also a driver-facing camera to make sure you're looking at the road on Autopilot/FSD.

Yes, there are massive data transfers and folks have noted about it on the internet. Just Google "[my tesla uploaded gigabytes](#)"

Subject	Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/01/2024 12:47 PM
Member	zax382

I'm sorry, I've sat this out thusfar but I just had to weigh in on this statement: Tesla can solve autonomy with vision only. They can't.

I've always been an auto guy and spent a lot of time on autonomy tech in the 2017-2020 timeframe. At the risk of doxxing myself I turned down a role running biz dev for a \$100bn+ company's autonomy division. As part of this process I spent weeks with the scientists and engineers central to pushing this effort forward, with extensive discussions on the various technology stacks required. The main reason I turned the job down was that it became clear that this truly is a science project, there are some extremely difficult computational problems here where no one is making a lot of headway excluding maybe Waymo and Cruise, and particularly not Tesla. They are in last place of the major players and will stay there because of a choice Elon made about his hardware stack 8 years ago. He was running out of money and so could not afford extremely expensive Lidar systems, so just asserted to his engineering team that they were not necessary and they have tried to push forward without them. You can have a pretty solid Level 2/3 product with vision only, and there are a couple good ones out there including FSD. But, long-term this is where Elon kills you – he's not really an in-the-weeds engineer and doesn't take no for an answer, so even when he's dead wrong on something (cybertruck, anyone), there's no one around to course correct. He's doomed their autonomy plans to failure. Let's talk about why!

1) Snow: Why does everyone ignore this. If you want true robotaxi service it has to be able to work in the snow, or sleet, or hail etc. Unless you want to permanently confine yourself to only 30% of the population of the US, you have to have a system that can work when there is blowing snow and lines are not visible on the road. Vision SUCKS at this. Cameras are easily blocked (they don't move, after all), especially those low to the ground, and you don't have the correct road perspective and GPS circling to define the correct lines. Vision sensors will see a line of snow and think its a white line and start changing lanes – its a disaster. And there is no programming your way around this as you just flat out don't have the data. Moreover, you are going to get a ton of misleading data from Tesla bc they are rarely driven in the snow and no one turns on FSD in bad snow because, well, its suicide. Until TSLA adds Lidar you will have no snow solution and no robotaxi solution.

2) Lidar's superior vectors – this manifests in two ways, looking down from the top of the vehicle and looking "around corners." The way the vision sensors work and where they are mounted its very easy for them to miss parallel traffic that is blocked by another similarly sized car/horizontal blockage. Its also hard to determine subtle lane change dynamics (imagine a lane moving over 4 feet in a construction zone, lidar sees this well ahead of traditional vision). Vision can get unlucky on blockages and just flat out miss cars moving perpendicular in tight spaces, things lidar will more often catch. Remember – a Level 5 autonomy robotaxi solution must be *perfect* given the number of reps its going to get. Doesn't work if it only kills the passenger on one out of 1 million miles, so the corner cases are all that matters here

3) Bright Flashing Lights: Very bright lights blind even really good vision sensors, but Lidar has no issue with these given the specific frequencies on which it operates and the rotary nature of the sensor. This is why Tesla's on FSD tend to run into parked firetrucks at 70mph. This is a really hard problem to solve with vision alone because in order to make the sensors subtle enough to see unlit objects at night (see below), really bright light blinds them in the other direction. You just can't do it without lidar

4) Hitting a deer: Lidar sees the deer. Vision misses the deer. Deer small. Deer unlit. Also insert: young child at night.

There's a reason everyone else uses a giant lidar suite. There are only two companies with the hardware/software stack that can offer full autonomy at any level right now, Waymo and Cruise. And its tough – Waymo is doing it in the flattest, best weather environment they can find in Phoenix and its taking years to just roll out additional geofenced areas, and Cruise tried to tackle something harder (SF) and has kinda failed to date. This a a really hard problem.

The path to fully autonomous autos does not pass through FSD, or any level 2/3 solution. It requires an extremely sophisticated hardware and software stack and millions/billions of miles of detailed testing.

One more word on the "data" – The idea that a bunch of bros out here in cars labelling "bugs" is some sort of advantage misunderstands the nature of what a "bug" is – drivers should not NOTICE most bugs. The real bugs are when various systems in the car contradict – vision sees X, Lidar sees Y, etc. If your "bugs" are the car like trying to change lanes into a truck, you should never have pushed that product. The disconnects that happen with other products are much more subtle and generally are times when the car just doesn't know what to do so it slows to a stop. Not trying to careen through a stop sign. You need to get to a place where people don't notice much other than your car being cautious before the product gets competitive, and at that point your bug identifiers are no longer useful.

Finally, without lidar your data is MUCH less robust to even do testing on. How do you program around the snow problem with a blinded camera? You don't have any inputs to even use to fix the problem. You need multiple data inputs with extremely robust series of data to even start. Tesla will need to redesign its hardware stack if it wants to push beyond Level 3.

Now, in reality, I think they know this – and will just push a level 3 product that's fine, if way more dangerous than something any other OEM would consider, and try to use it to pump the ultimate product – which is the stock.

Subject	Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/01/2024 01:48 PM
Member	cuyler1903

Well, that's one of the best message posts in VIC history right there.

Nice.

Subject	Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
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Entry
Member

05/01/2024 02:11 PM
WinBrun

Thanks for those interesting insights. So assuming you're right about Lidar being the necessary piece of the puzzle and Tesla ultimately reaches that conclusion, do you think they would incorporate it and use a vision + lidar hybrid solution? Or would they throw in the towel on autonomy? And if they would incorporate it, what sort of challenges would they face in terms of tech, costs, or otherwise?

Subject
Entry
Member

Issue Tree
05/01/2024 02:15 PM
bowd57

Hi, all –

WMC, by way of background, I'm short Tesla in size. Shorting a cult stock in size is probably not a good idea ever, but here we are. In theory I should be open to, not just covering, but going long, but my views on Musk are so negative that I don't see that happening under current management. Since this is not my writeup (Yay!), I don't need to argue for the short. And I'm not going to. I'm not trying to convince anyone of anything. What I'm hoping to get out of the conversation is ... background, thoughts, vibes, perspectives, to help me with my "short cult stock in size" problem, in particular, should I cover? I hope the rest of the club will find the discussion useful or interesting.

I think bulls and bears generally agree that autonomy is the main value driver.

Here's my ... I'm not sure what this kind of thing is called, but I'm going with "Issue Tree":

1 Are driverless taxis possible?	
2	Yes, they exist in multiple cities.
3 Can a driverless taxi service be profitable?	
4	You'd think, but it hasn't been done yet.
5 Can Tesla ever hit the required MTBF?	
6 If so, how long might that take?	
7	How far away are they?
8	How rapidly might they progress?
9 Can they do it with the current fleet?	
10	Yes
11	Requires modest but time-consuming retrofits
12	Not at all
13 Costs/timeframe to roll out service? Regulatory, operational, etc.	
14 What about the competition?	
15 Tesla potentially has this "personal chauffeur" angle that's worth considering.	
16 What about really, really capable L2?	
17 What might RT/PC be worth?	

In the short term, which, as a "short cult stock in size" guy is very important to me, I think #7 and #8 are the crucial issues; like, I might (not saying I do) personally believe that ultimately robotaxis are valueless, but right now the market sure doesn't. Here, courtesy of the log function, is some sort of sensitivity analysis of #7 & #8:

MTBF X needed	10000		
MTBF X rate	10	4	2
Years to goal	4	7	13
MTBF X needed	1000		
MTBF X rate	10	4	2
Years to goal	3	5	10
MTBF X needed	100		
MTBF X rate	10	4	2
Years to goal	2	3	7
MTBF X needed	10		
MTBF X rate	10	4	2
Years to goal	1	2	3

, where "MTBF X" means, "MTBF times", so "1000x MTBF needed" means MTBF needs to get 1000 times better, "4x MTBF rate" means MTBF is increasing 4x annually. My narrative for the "MTBF needed" levels is: 10000x = "Superhuman", 1000x = "Human", 100x = "The tracker is really off", 10x = "You're high".

WMC, JGalt, and the club as a whole -- Does this make sense? Am I way off-base? Am I thinking about this wrong?

Assuming I'm not just sniffing glue, a question for the longs is: What does 4x annual MTBF improvement mean for? I assume that at 2x, you'd bail out, I don't think anybody's signing up for a 7-13 year long development cycle. If I believed or could see 10x, I'd probably bail, I just wouldn't want to be short whatever my long term views are. It's that # in the middle that's troubling. If you became convinced that RTs were 5 years away, would you say, "I'm out," or "Disappointing but I'm sticking with this?"

Unfortunately, the crucial questions here require a crystal ball to answer, we're all just guessing. WMC, let me run my understanding of state of play, the so limited as to be practically worthless data we've gotten to date, by you to see I'm off base. Prior to 12.x, the FSD program was a joke that made no progress at all. The raw #s from 12.x are still a joke, but (A) there *is* progress, and (B) the claim is that Tesla has set up a machine to reliably generate further progress. The tracker tells me that 12.x is 2x better than prior versions after a years development. That's really hard to interpret or extrapolate from, though, because on the one hand, "In development", and on the other, "Low hanging fruit." FSD 12 has now been in the public's hands for all of *two months*, an incredibly short time. But the 10x number I'm worried about means +78%/quarter, +22%/month, and that hasn't happened yet.

Yours,

Bowd

Subject	Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/01/2024 02:20 PM
Member	katana

zax: Please stop posting this garbage and clogging up the home page with useless TSLA board posts

Cuyler: *Obviously*. (Translation: If you're also reading Bowd's TSLA thread, you know that my first sentence to zax is 100% sarcasm and a dig at a certain TSLA-message-board hater)

WinBrun: *"Or would they throw in the towel on autonomy?"* They cannot possibly throw in the towel on autonomy; it would be conceding the death of the business. This is their Stalingrad, if it comes to that.

Subject	Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/01/2024 02:21 PM
Member	bowd57

Hi, Zax --

I agree about the end state. The market doesn't. Would you care to apply your vastly superior insights to the question of, "Can Tesla fake enough progress over the short term to make the stock a buy/cover over the short term?"

Yours,

Bowd

Subject	The Simplest Way To Think About Sensors
Entry	05/01/2024 02:24 PM
Member	bowd57

Hi, all --

Apropos of Zax's post:

Musk talks about the march of nines. At some point along the way, radar turns into a nine. At some point along the way, lidar turns into a nine.

Yours,

Bowd

Subject	For zax382
Entry	05/01/2024 03:09 PM
Member	jpgalt

Best retort to your post is: "humans can drive with two cameras on a gimbal, therefore, so can cars"

This is Elon's argument and it makes sense.

"Cameras get dirty" OK then, let's include a way to clean the cameras, this is not hard to solve.

Cars with cameras can be much safer than humans because they don't get distracted, don't drink, don't fall asleep at the wheel, etc.

Subject	Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/01/2024 03:10 PM
Member	zax382

No, I don't think they throw the towel. I think they will go as far as they can go with vision (FSD enhancements), and then will debut a lidar-related solution to get to Level 5. I just think its necessary. I also wouldn't be surprised if third party Lidar suites start emerging, I do think there have been many comments about how many players will just adopt the best-in-class solution vs their own development. I wouldn't be shocked to see a TSLA-Waymo partnership in 2-3 years.

I just think this will take a while because the bottom line is -- no one is particularly close to a Level 5 robotaxi solution. This is 2030's we are talking about here (unless there have been breakthroughs at Waymo, the leading player, that I don't know about), so there's little incentive / reason for Elon to change course because its not like there's going to be some awesome product that debuts near-term that shows how far behind they are.

My belief on the way robotaxi's roll out is on a geofenced basis, so Waymo goes to 5 more cities, maybe Cruise tries a different few cities, each city takes 6-12 months, and by 2030 maybe you have 30 cities where you can hail a robotaxi. But, still geofenced pretty aggressively, and probably not running in the rain or snow. Of course this doesn't comport with some Tesla that can just drive anywhere -- that challenge is so hard!

But in the near term I don't anticipate Elon changing course because he can just keep promising something that is "medium term" that never comes, and doesn't have to actually deliver until someone else does, which is gonna take a while.

Subject	Re: For zax382
Entry	05/01/2024 03:12 PM
Member	zax382

Humans wreck cars ALL THE TIME. 35,000 people die in car accidents every year in the US. Some of those are because they don't see stuff coming. The problem is, your robotaxi can't kill *anyone*. It needs to be way better than two cameras on a gimbal.

Subject Re: Re: For zax382
Entry 05/01/2024 03:33 PM
Member jgalt

I'm aware of the stats. Still think it'll be way safer than humans with vision only. We'll see!

Subject Another Anecdote
Entry 05/01/2024 03:55 PM
Member Wrangler

I'm a Full Self Driving skeptic, and admittedly hadn't tried v12.3 yet, but based on compound's anecdote I decided to borrow a Tesla and try it out today. I did a 26 mile route with 5 stops, total drive time 67 minutes. I would describe the route as ideal weather, ideal traffic conditions, ideal roads. When it works, it's amazing. But it doesn't always work...

The errors it made in that short trip:

3 traffic violations that could have required critical disengagements if other cars were around, but I just let it do its thing because no cars were around. 1 it was in the right turn lane where it intended to go straight. Probably annoyed the person behind me also trying to right turn, and nobody was in the lane to the left which is the lane that is supposed to go straight, so if someone was in that lane, I don't know if the Tesla would have just cut them off, or if it would have acted differently. Same situation happened in another instance but in this case the one left turn lane was so obvious, separated by a large white dashed buffer lane from the straight lanes, and it put on its blinker, acted indecisive (on assertive mode) about which lane to choose, then ultimately chose the straight lane but still turned from that lane. Again, I let it do it since no car was in the left turn lane but I'm not sure if it would have been a critical disengagement if there was traffic. The third instance there was a dead end and a left or right turn lane, it got in the right turn lane to turn right (where it was mapped to go), but then randomly decided to turn left sharply. Again, no traffic, so I didn't disengage and nobody got hurt.

In and out of my guard gate to my neighborhood it had no intent to stop for the gate arm, so I had to disengage or crash into the gate arm (twice).

I routed it through 3 school zones and it didn't recognize any of them so it was going 40mph through a 15mph school zone until I disengaged. I'm not sure why it can't recognize school zones when there are flashing yellow lights on a "school zone" sign.

In one of the school zones, I disengaged because a stop sign holder walked out into the middle of the street from behind a car where she wasn't visible. I can't know if it would have run her over, but it sure felt like it was going to.

In one instance on the highway, it merged at 60mph in front of an accelerating truck where the truck and normal traffic flow were closer to 75mph. Felt like we would get rear ended so I disengaged to accelerate.

Overall, I was impressed that it got me from point a to point b without dying and without intervention (though with traffic violations that could have killed me if other cars were present) on 4 out of the 5 legs of my 5 leg route. BUT, using Bowd's favorite tool, the tracker, I shouldn't expect to have any critical disengagements on a 26 mile route. I need to do that ~10 more times to expect a near-death experience.

Jgalt & Weighing Machine, are your experiences similar and you just think it'll work out those "kinks" with the feedback loop of driver disengagements? Or are your experiences more flawless?

Subject Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry 05/01/2024 04:12 PM
Member Motherlode

Interesting. wouldn't you add rain to your list? Its extremely challenging to back up my TSLA when its raining bc the rearview window is tiny and the camera is useless.

Subject Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry 05/01/2024 09:33 PM
Member Weighing Machine Capital

@zax382,

Great points you make about LiDAR and I completely agree with you that Tesla couldn't afford expensive LiDAR in 2016 and that is why they left it out. They would add LiDAR down the road when it got much cheaper if pure vision couldn't solve autonomy on its own. Or at least add other sensors such as HD radar to the front bumper which they are doing now.

A few questions:

- 1) Can LiDAR solve the snow issue and if so, how?
- 2) Can you explain how LiDAR looks "around corners"?

I'm surprised you didn't mention anything about HD maps re: Waymo and Cruise. Arguably, that's a bigger component of the robustness of their systems than LiDAR. (Tesla doesn't use HD maps either.)

- 3) What is your professional background/experience in?

I have no background in computer vision/AI. I have only learned what I know, or think I know, from spending way too much on machine learning/signal processing/computer vision experts.

Thanks for your insights.

Subject	Odds vs Edge?
Entry	05/02/2024 09:27 AM
Member	nha855

What do you think you're playing for here? 5x? 10x? If FSD ends up not being viable, how much do you lose? 100%?

How do you see the odds that FSD ends up working and being the industry leader? 50/50? Better?

Subject	Re: Re: Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/02/2024 09:39 AM
Member	zax382

Sure, I can help on these:

1) The snow issue is a real issue, and not one where Lidar is some killer app. It can be better in the snow for a few reasons. First, as you know, Lidar systems that, for example, sit on top of the car spin rapidly sending out pulses of light (900nm although now 1400nm is top of the line) and whatever it hits you get data on its location in three dimensional space. So, in a snowy environment, lidar is going to throw back all sorts of hits, but the algorithm for screening those hits is going to be pretty good – snowflakes are small and move, and so your constant data feed showing the flake 6 feet off the ground and its a total of 3mm tall, well its easy for your system to know that is a snowflake. Now, if you add multiple Lidar units this gets even better because you are able to construct the 3D world from multiple angles and identifying a snowflake is much easier. Second is just that blocking the sensor is much harder, both because of where and how it is mounted, but also because its sending out light at 360 degrees spinning very rapidly, vs a fixed camera. Finally, and this is more technical and I cannot promise to be an expert, with new Lidar systems in the 1400nm+ range, I think there is also some ability to "see through" the snow, or at least have such precision that you can tell the difference in snow scattering vs solid object. Don't quote me on this one though. Snow is still tough, especially for identifying things like road lines, which is where (as you mention) HD maps and GPS become critical.

2) Sure, this is due to how its mounted and the 3D nature of how it constructs the cars. Simple example, you are at a stoplight, car next to you on the right, light goes green you pull out, car in intersection runs red light coming towards you perpendicularly. Your top mounted lidar sensor has a much better chance of seeing that car coming a ways away, while a vision system could just straight up be blocked depending on the circumstance and not see the car until its too late.

3) I'm a long-term autos investor, as I said on my first post spent a lot of time on autonomy when considering a career change into the industry.

Subject	Re: Re: Re: Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/02/2024 10:40 AM
Member	yxd0950

Hi Zax,

here's one dumb question which was also raised by someone else: why can't we just have something wipe off the snow on the camera?

Thanks.

Subject	Re: Re: Re: Re: Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/02/2024 11:15 AM
Member	zax382

You can, it just:

- 1) Adds a mechanical piece that can easily break
- 2) Takes time – imagine in constant snow where camera is blocked every 30 seconds. Its just not super reliable
- 3) Still doesn't solve the key issue which is where the camera is mounted

Would be interesting to see this added to new TSLA's though -- haven't seen anything like it really other than a few backup cameras that have water jets

Subject Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry 05/02/2024 11:17 AM
Member bowd57
 Hi, Yxd --

> why can't we just have something wipe off the snow on the camera?

You could but they don't. I regard this is being part of line 11 in my "Issue Tree", outlined in message #24, which I wish everyone would use, or at least use after improving upon, in order to enable a more disciplined conversation.

Yours,

Bowd

Subject MBLY - the much cheaper and more attractive way to play Autonomous Driving
Entry 05/02/2024 11:34 AM
Member jso1123

Sorry to interject but you guys are all missing the much more attractive and straightforward way to play autonomous driving and you can buy it for <\$25bn market cap: MBLY.

High level:

MBLY is the leader in autonomous driving features with better performance than TSLA (MBLY is at 100 / 500 MTBF with EyeQ5 / EyeQ6 chip vs. TSLA at 5-10 MTBF; why don't people talk more about this fact?). MBLY's L2+ "SuperVision" is a very good product.

MBLY's TAM opportunity is the 70%+ of the automotive market that isn't TSLA, a handful of Chinese OEMs, and Mercedes/BMW doing it themselves, so 60mm+ cars vs. TSLA's TAM is 2mm cars today going to (maybe?) 5-10mm cars over time. We think it is unlikely TSLA has meaningful success licensing FSD to other OEMS (TSLA has proven they are an unreliable partner; look at their SuperCharging network announcement; also, OEMs fear TSLA would have supreme pricing power over them if licensed FSD so would not put themselves at existential risk by outsourcing one of if not the most critical and differentiating functions of a car in 3-5 years to their direct competitor).

L3/L4/L5 "eyes off" autonomy (including robotaxis) is a long way away due to the exponentially increasing challenges of solving for edge use cases and the very high safety bar required for these systems (need 100k-1mm MTBF vs. human drivers crashing around ~75k MTBF). L2+ will be the sweet spot for adoption of autonomous-like features over the next 3-5 years and MBLY provides a scalable path for OEMs to go to L2+ and then L3/beyond as these features mature. OEMs have been trying to design their L2+/L3 systems in-house running atop NVDA or QCOM hardware but are increasingly hitting a wall due to the challenges of developing the technology and we believe the pendulum is shifting back toward MBLY. MBLY is the only commercial solution available that also is the cheapest price point (\$1.5k for SV).

MBLY uses a camera + lidar/radar system that solves for the issues with a camera-only system.

Importantly, MBLY is currently engaged in RFQs with 14 OEMs equating to 46% (almost half) of global SAAR. MBLY's win rate on engagements thus far has been >70%. We think MBLY can secure a number of large design wins for L2+ "SuperVision" over the next 12 months that will validate their competitive position and long-term TAM opportunity. Stock is \$30 today and think they can do \$4-5 of long-term EPS.

Worth a look. Recommend reading the Wolfe note from today - does a good job of explaining MBLY.

Subject Re: Re: Re: Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry 05/02/2024 12:05 PM
Member Weighing Machine Capital

@zax382,

Tesla can add a camera mounted to the top of the car to solve the problem you describe of adjacent cars blocking the view of perpendicular traffic. The solution is the top of car mount to see over adjacent objects, not that it is LiDAR vs. camera.

In your original post, you wrote that LiDAR looks "around corners". Can you elaborate how it does this?

Thanks

Subject	Re: MBLY - the much cheaper and more attractive way to play Autonomous Driving
Entry	05/02/2024 12:07 PM
Member	Weighing Machine Capital

@jso1123,

Have you personally used MobileEye's SuperVision or Tesla's FSD v. 12.x?

Subject	Re: Odds vs Edge?
Entry	05/02/2024 12:27 PM
Member	Weighing Machine Capital

@nha855,

If FSD doesn't solve autonomy, Tesla is down > 75%

If FSD solves autonomy, and the 2nd place player is a distant 2nd place, then Tesla > 10x. (A lot of people buy a Tesla car to get FSD)

FSD is the only system architected the right way to solve autonomy and will distance itself from others as network effects kick in as usage increases.

It's intentionally an L2 system; Tesla needs the humans supervising it.

The bros driving around disengaging FSD is the key to the march of nines.

Subject	Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/02/2024 12:49 PM
Member	finn520

What percentage of U.S. miles driven in 2023 were while it was snowing? 1%? If it doesn't work in the snow, can't people just drive normally then and have FSD the other 99% of the time. Similar thing with rain although number is obviously higher, though problem doesn't seem as hard.

Subject	Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/02/2024 01:04 PM
Member	Weighing Machine Capital

@finn520,

Exactly.

FSD uses **imitation learning**. Tesla has plenty of data of snow driving. FSD doesn't need clear lane lines. It uses drivable space when lane lines are messed up; just like humans do.

Someone wrote that snow is an issue for FSD because people don't use FSD in the snow and thus, no snow driving data. Doesn't matter if FSD is engaged in the snow; the human driving data is sent up regardless.

Subject	Re: Re: Odds vs Edge?
Entry	05/02/2024 01:24 PM
Member	nha855

Thank you. My impression is that you think it's > 50% that TSLA has a strong first place whereas skeptics see it as < 10%

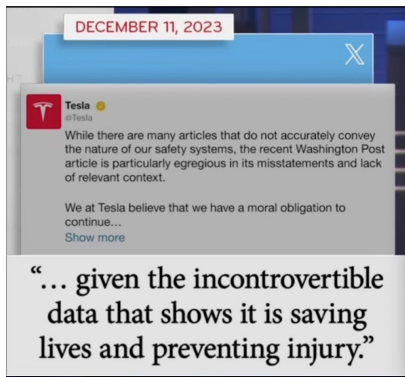
Subject	Re: Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/03/2024 06:40 PM
Member	JWF211

You say FSD is 99% the way there... yet all the data shows its so, so far away from approaching human-level, let alone super-human, driving.

A commercial-grade robotaxi service would need to be significantly safer than a human driver. FSD needs to improve 1,000-2,000x before reaching a Waymo-like safety profile, for example. On what basis do you believe there is a credible path for Tesla to do so?

The idea that the advent of a transformer model somehow bridge this gap seems silly.

Subject	Re: Re: Re: Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	05/06/2024 09:09 PM
Member	SlackTide



NHTSA claims that limiting to this level means Tesla is only collecting data on around 18% of crashes that are reported to the police. As a result, NHTSA wrote that the probe uncovered crashes for which Autopilot was engaged that Tesla was not notified of via telematics.

Thoughts?

Subject
Entry
Member
Hi, SlackTide --

Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
05/06/2024 11:15 PM
bowd57

I don't want to say that Tesla isn't serious about safety, but they're absolutely not serious when communicating about safety. Tesla's self-reported stats can't be taken at face value. They under-communicate and under-adjust and over-redact every chance they get.

Because Tesla isn't serious about communicating about safety, there's not a lot to definitively said. There's a non-Tesla-war researcher out there (from Cornell?) who says that, after making some obvious adjustments, Tesla's alleged safety advantage vanishes, and that making a few more obvious adjustments turns them into the red, which is kind of what you'd expect from cars that provide EV acceleration. I'm too lazy and undisciplined to have pointers to this available, if there's interest I'll try to dig it up. Insurance data shows that Teslas are 50% more likely to be involved in a crash *per owner*, note that's not per mile, again I could dig this up.

The "18% of crashes" sounds really bad and is really bad, but at least 60% (I made that up) of that is "communicating about safety" bad as opposed to bad-bad.

Really briefly: Teslas are new cars with the latest safety features driven by wealthy middle-aged people which is going to skew accidents down, but then have incredible acceleration which will skew things up. Claims that Autopilot (NB that's not FSD) or similar systems improve safety should be regarded with a lot of suspicion, because, hey you know what, they weren't designed with the intent of improving safety to begin with. It's incontrovertibly the case that people have placed undue reliance on Tesla's automation systems. The role of the CEO in this is a subject of active and vitriolic debate.

Yours,

Bowd

Subject
Entry
Member

Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
05/07/2024 12:24 PM
GCA

"Insurance data shows that Teslas are 50% more likely to be involved in a crash *per owner*"

Incredible

Subject
Entry
Member

Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
05/07/2024 01:31 PM
bowd57

Hi, all –

Me:> There's a non-Tesla-war researcher out there [...]

This guy:

<https://profile.virginia.edu/njg2q>

<https://twitter.com/NoahGoodall>

He should really see if he can get a role in a vampire movie.

Yours,

Bowd

Subject	where the hell is bowd? this is your moment
Entry	05/08/2024 08:52 AM
Member	mojoris



Carl Quintanilla  @carlquintanilla · 47m

* EXCLUSIVE -- U.S. PROSECUTORS EXAMINING WHETHER TESLA COMMITTED SECURITIES, WIRE FRAUD IN CRIMINAL PROBE OF SELF-DRIVING CLAIMS, THREE SOURCES SAY

[@reuters](#) [\\$TSLA](#)

 36
 189
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Subject	Re: where the hell is bowd? this is your moment
Entry	05/08/2024 09:04 AM
Member	bowd57

Hi, Mojoris --

I don't regard this as news in the light of Tesla's disclosures of prior subpoenas and the, well, the obvious.

Yours,

Bowd

Subject	Re: Re: where the hell is bowd? this is your moment
Entry	05/08/2024 10:04 AM
Member	katana



Subject	Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	07/29/2024 10:16 PM
Member	natey1015

https://www.wsj.com/video/series/tesla-autopilot/the-hidden-autopilot-data-that-reveals-why-teslas-crash/68D26569-0251-4637-A035-A5131D8883B8?mod=prem_login_68D26569-0251-4637-A035-A5131D8883B8#video-player

Subject	Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	07/30/2024 12:16 AM
Member	bowd57

Hi, Natey --

Care to summarize for the commies among us without a WSJ sub?

Yours,

Bowd

Subject	Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	07/30/2024 02:09 AM
Member	bowd57

Hi, all –

I wandered across the video Natey links to, and there's little new to those who have been following.

Yours,

Bowd

Subject	Re: Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	11/19/2024 11:28 AM
Member	Weighing Machine Capital

When FSD version 13 is released right before Thanksgiving, the world is finally going to wake up to what Tesla has been building for the past decade...

Subject	Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	11/19/2024 12:10 PM
Member	bowd57

Hi, WMC --

Eagerly awaiting! Congrats, by the way.

Yours,

Bowd

Subject	Recent rally based on stability of Trump - Musk relationship - not likely to last
Entry	11/19/2024 01:36 PM
Member	Paradox

I am enjoying playing around with AI. I constantly ask silly questions like "Why is Tesla stock up so much in the past 6 months?"

Here's the answer:

Based on the information provided, Tesla's stock (TSLA) has seen significant growth over the past 6 months due to several key factors:

1. **Strong Financial Performance:** Tesla's stock price has risen from around \$170 in May 2024 to over \$345 in November 2024, representing a gain of approximately 103% over the 6-month period ¹ ².
2. **Trump's Re-election:** The re-election of Donald Trump as president has been a major catalyst for Tesla's stock surge. Since the November election, Tesla's share price has gained 28%, outperforming broader market indices ⁴.
3. **Regulatory Tailwinds:** The anticipated regulatory changes under the incoming Trump administration are seen as favorable for Tesla. There's a proposed federal framework for self-driving vehicles that could streamline development and testing processes. Additionally, potential easing of corporate tax regulations and tariffs on EV components has boosted investor confidence ⁴.
4. **Musk's Influence:** Elon Musk's significant contributions to Trump's campaign (over \$100 million) and his emergence as a top advisor in Trump's inner circle are seen as beneficial for Tesla. This "strategic alliance" is expected to create a more favorable regulatory environment for Tesla's autonomous driving ambitions ³.
5. **Autonomous Driving Potential:** The prospect of a federal framework for fully autonomous cars is seen as a major opportunity for Tesla. Some analysts estimate that autonomous driving alone represents a \$1 trillion opportunity for the company ³.

6. **Product Innovation:** Tesla continues to lead in product innovation, with announcements like the Cybercab autonomous vehicle concept generating significant market interest, even though production isn't expected until 2027 ⁴.

7. **Financial Initiatives:** Tesla has announced a \$1.25 billion share buyback program for fiscal year 2025 and increased its dividend by 4.5% for FY24, signaling financial strength and commitment to shareholder value ⁴.

It's important to note that while Tesla's stock has surged, many analysts consider it overvalued. The average analyst price target is \$207, which is 33% below the current trading price ⁵. The stock's valuation reflects high expectations for future growth in areas like autonomous driving and robotics, rather than just its current performance as an EV manufacturer.

Then it struck me that most of this run up is based on the perception that Trump and Musk are very well aligned so I asked the next question:

"How long did the average cabinet member last during the first Trump administration?"

Answer: 7 months

It seems to me that a huge rally based on the relationship of two less than stable guys with a history of tension isn't likely to last very long.

Subject	Re: TESLA SECRET SAUCE TO SOLVE AUTONOMY
Entry	11/28/2024 04:41 PM
Member	bowd57

Hi, WMC --

I'm keeping my eyes peeled!

Yours,

Bowd

Subject	Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 09:52 AM
Member	Weighing Machine Capital

I just experienced FSD version 13.2. Holy Shit.

AUTONOMY IS SOLVED.

I AM 100% CONFIDENT THIS IS TESLA'S iPhone MOMENT.

EVERYONE WILL ORDER A TESLA KICKING AND SCREAMING WITHIN 6 MONTHS TO GET ACCESS TO FSD.

Subject Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/05/2024 10:08 AM
Member heffer504

It's very fun to watch technology advance, for sure. You should try it on a rainy night and see how it goes... But I would rather eat glass than support that piece of sh*t. I can drive my own car...

Regards, Heffer

Subject Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/05/2024 11:46 AM
Member natey1015

What happens to FSD when cameras are not clear due to rain, fog, snow, glare, mud, dirt, etc.? If cameras are not 100% clear all the time can a Tesla ever be fully autonomous akin to a Waymo (which uses Lidar, radar and cameras) given it relies solely on cameras and sensors?

I am on my second Tesla Model Y. I'm not a car guy and it's the first car I've ever loved so I am not a Tesla hater. Full disclosure: I don't own the stock.

I love using Waymo too. But I've never had any interest in using FSD (even when Tesla has offered me free monthly trials) because it's not a full autonomy solution. What's the point of using FSD if I have to keep my hands on or close to the wheel at all times and my eyes on the road most of the time?

With Waymo I can completely tune out, never look up from my phone while reading/writing emails or text messages, etc. Bottom line I would not put mine or my family's lives at risk using FSD in a Tesla, believing it's fully autonomous and acting accordingly when driving. On the flip side, I'm very comfortable with family in a Waymo--particularly given the data showing how safe the cars are and how it has worked with regulators to prove just that.

Subject Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/05/2024 01:16 PM
Member Weighing Machine Capital

When researching an investment, we try to sift through as much information/data we can; eliminating 99% of it as noise, to get to the 1% that is signal.

It shocks me that the investment community relies on misinformation from the combustion automobile industry, and <https://teslafsdtracker.com/> (which is complete garbage in/garbage out) and is irrelevant as it's not about the absolute capability of FSD; it's about the architecture and progress that matters.

We all can experience the progression of FSD in real-time by using it and objectively judge for ourselves.

Autonomous transportation will be the biggest technological advance mankind has ever seen.

You really should try FSD version 13.2 when it goes wide release in the next few weeks to experience it for yourself.

Subject Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/05/2024 01:48 PM
Member JohnKimble

I have a 3 and an X and use them to commute into midtown manhattan from NJ. FSD is a game changer for anything from the tunnel outward, but it doesn't work in Manhattan yet. It also annoyingly still wants your occasional attention even in slow traffic. In the 3, you can actually put the sun visor thing in front of the inside camera and then scroll twitter while your hand is on the wheel, but the X has a tiny sun visor thing and seems to be on to my games. Sunglasses work, but you still need occasional hands on the wheel.

Merging in heavy traffic isn't solved yet, toll booths aren't solved yet, Manhattan isn't solved yet, and the phantom braking thing still happens to me sometimes. It's totally blind to potholes, and some of the potholes in the tri-state area could engulf your vehicle. Once, it braked hard for a truck that was *transporting traffic lights*. I jokingly wonder if there are so many edge cases that you basically need AGI to make FSD work everywhere.

With all that said, FSD has gotten a LOT better in the past year. It works almost flawlessly in country/suburb settings without the psychopathic NY commuters, and it does feel like its close to being ready.

Subject Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/05/2024 01:48 PM
Member T0YPAJ182

I have had FSD since 2020. I for sure deserve to be called a sucker, at least up until this year. I agree the progress has been on a different level since the architecture overhaul, and TSLA equity has justifiably reflected that. Between Waymo in the immediate-term and Tesla medium to long-term, do you have any view on Lyft or Uber shorts? I feel like we will look back in five to ten years and point to the time of autonomy accelerating and be like, "UBERQ and LYFTQ really were layup shorts."

Subject Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/05/2024 02:52 PM

Member

heffer504

"Autonomous transportation will be the biggest technological advance mankind has ever seen"

I really think you need to question whether you are able to make intellectually-honest decisions after a statement like that. It seems impossible to me that any sane person could believe that statement. So, if you do truly believe that, I would suggest that you find an arm's length third party to evaluate your position sizing on this one.

And, to paraphrase my prior statement, sewer rat may taste like pumpkin pie, but I wouldn't know because I wouldn't eat the filthy motherfucker.

Good luck. Heffer

Subject
Entry
Member

Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
12/05/2024 03:15 PM
T0YPAJ182

"It seems impossible to me that any sane person could believe that statement."

Have you considered that it is perhaps you that is not sane? Weighing Machine is making statements about firsthand experiences with the product over time (not to mention it has been an incredible call) while you make emotionally-charged rants about something you have no firsthand experience with as a consumer.

Subject
Entry
Member

Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
12/05/2024 03:23 PM
Paradox

Pretty basic question here...but met my new neighbor this morning and apparently he owns the local GM dealership. He drives an EV Hummer....looks pretty cool to me so I asked him about it. He bluntly told me that the crab walk is marketing hype and a bit silly as its decent for four wheeling but Hummer buyers are actually high income luxury car types...but he went on to say the FSD mode (cruise) is incredible and that he just took it from San Diego to LA (100 miles) and it was on FSD for all but 5 miles of the drive.....anecdotal evidence not generally the best kind of evidence...anyway....

Competitive Advantage & First Mover Advantage question: With at least 5 major well funded players in this area of technology aren't most car companies likely to have this technology in the future? The history of technology in cars is all companies get around to including it and it never amounts to much of an advantage for very long.

Is it going to be different this time?

Subject
Entry
Member

Grumble
12/05/2024 03:35 PM
bowd57

Hi, all –

I may be back with more in a bit, but I did want to hammer home two points:

- 1: The debate is about reliability, not capability. I can ride a bicycle with my eyes closed! Neat trick, huh? The question is, for how long? And that's why,
- 2: WMC's claim that,

WMC:> We all can experience the progression of FSD in real-time by using it and objectively judge for ourselves.

, is completely off-base. This is a matter of stats, not personal experience.

Yours,

Bowd

Subject
Entry
Member

Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
12/05/2024 04:13 PM
heffer504

I haven't laughed this loud in a long time, thank you! And of course I have driven in a Tesla, I just wouldn't buy it because I have some shred of morality despite working in finance.

But OK, you seem to be a serious person and clearly agree with the statement so I will engage you seriously (but with a smile on my face, I can't help it).

I invite you and Weighing Machine to no longer use computers, the internet, penicillin, anaesthesia, airplanes, GMO-based food products, shit I don't know pick a million technological advances that are clearly more important than FSD.

Just give up one. Come one, just... let's say airplanes. You can never fly again, but you can own a Tesla with FSD. That should be an easy decision, right? Promise me right now you will never fly again, and I promise I will never drive in a Tesla with FSD. Pinky promise?

Subject	Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 04:48 PM
Member	T0YPAJ182

Deal. But only if a Joby S4 is excluded from airplanes ;)

Subject	Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 05:47 PM
Member	cuyler1903

JohnKimble, I think I've identified your primary gripe.... It's not with FSD, but with living in NY/NJ! Get out of Dodge, quickstyle!

☆ 66	JohnKimble	4 hrs ago	Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED	↗ 65 🔒 🔗
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I have a 3 and an X and use them to commute into midtown manhattan from NJ. FSD is a game changer for anything from the tunnel outward, but it doesn't work in Manhattan yet. It also annoyingly still wants your occasional attention even in slow traffic. In the 3, you can actually put the sun visor thing in front of the inside camera and then scroll twitter while your hand is on the wheel, but the X has a tiny sun visor thing and seems to be on to my games. Sunglasses work, but you still need occasional hands on the wheel.

Merging in heavy traffic isn't solved yet, toll booths aren't solved yet, Manhattan isn't solved yet, and the phantom braking thing still happens to me sometimes. It's totally blind to potholes, and some of the potholes in the tri-state area could engulf your vehicle. Once, it braked hard for a truck that was *transporting traffic lights*. I jokingly wonder if there are so many edge cases that you basically need AGI to make FSD work everywhere.

With all that said, FSD has gotten a LOT better in the past year. It works almost flawlessly in country/suburb settings without the psychopathic NY commuters, and it does feel like its close to being ready.

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Subject	Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 07:16 PM
Member	heffer504

I agree to your terms

If you ever take a plane again, please don't lie about it, as that would be contrary to the spirit of VIC. I will be asking you every 3 months or so.

I assume you will be staying local for the holidays? Forever?

Meanwhile, feel free to ask me if I have bought a Tesla yet, as frequently as you would like

regards, heffer

Subject	Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 07:18 PM
Member	JohnKimble

On NY/NJ.... LOL fair

Subject	Musk
Entry	12/05/2024 08:01 PM
Member	bowd57

Hi, all –

Let me clear some underbrush before continuing.

There have been some heartwarming expressions of concern about my mental health. Some members fear that I'm obsessed with Elon Musk, that I suffer from some sort of derangement syndrome. But after discussions with my therapist, psychiatrist and parole officer, I think I'm OK.

Again, I appreciate the concern! It's true that Musk isn't my favorite CEO. In fact, I deeply resent him. It's because of him that I know who Amber Heard is, and that's knowledge I'd rather not have.

But I think it entirely possible to discuss Tesla, the business, without reference to Musk. Not the stock, of course; Musk moves the stock, he is the stock! And as we may see, it gets harder to leave him out of discussions of FSD. But if we can agree that he's had some hits and some misses, and that one should be very wary about making investment decisions based on his forward looking statements, then we can focus on the fundamentals of Tesla, the business, without getting into pointless debates about the difficulty of landing rocket ships on barges, or whether Mexican cartels were in fact with Musk's approval smuggling meth through tunnels to a Tesla warehouse.

Here, let me give an example.

Speaking of hits and misses, Tesla released an enormous hit in 2020, the Model Y. Around then, it announced a series of initiatives in fields ranging from heavy duty trucking to supercomputer design, raising hopes that company would continue move from hit to hit. Thus far, all have been misses, or to avoid needless quibbling, thus far the benefits of these initiatives have been unclear. And so we find ourselves with the value of the ?nth largest (I haven't checked the league tables lately) US traded stock almost entirely dependent on its ability to achieve autonomous driving in a commercially relevant timeframe. That's why need to focus on FSD, and why the subject is of interest even if you're very firmly in the, "No position, never, no way" camp. The subject is also of deep interest to those invested in related stocks, e.g., Uber.

See? No Musk! It's not that hard.

Give me a sec while I see if I can concisely steer the conversation in a more useful direction. Do note that none of this should be considered investment advice or any kind of forecast of where Tesla's stock might go. There are good reasons for me to not opine on those matters. This is a purely technical conversation.

Yours,

Bowd

Subject	Really, Really Basic Stuff
Entry	12/05/2024 08:13 PM
Member	bowd57

Hi, all –

I can't believe I have to do this.

It takes strong evidence to (statistically) prove a weak effect. Weak evidence can (statistically) disprove a strong effect.

I'm sure you've at some point eaten a bad peanut. Pervert that I am, I actually like them, as long as they're not too far gone. Say some promoter tries to sell a couple of trainloads of peanuts claiming that only one in a million is bad. Say you try one of them, and, guess what, it's bad. That is very weak evidence, and a very small sample size, but enough to (statistically) reject the claim that only one in a million is bad.

Supposed you're not allowed to sample the peanuts, but can visually inspect them. Say that in your experience (and mine!) bad peanuts are associated with discolored/broken/otherwise off shells. You observe that one in a hundred shells is visibly off. Personally, I'd at that point assume that 1/1000 peanuts were bad, just based on guessing that 1/10 of off shells contained a bad peanut. That guess could be way off! But I'd be really certain that more than 1/1MM peanuts was bad.

This is all of course a metaphor, but I'll save the punchline for the next post.

Yours,

Bowd

Subject	Peanuts & The Tracker
Entry	12/05/2024 08:15 PM
Member	bowd57

Hi, all –

Sorry, this got screwed up, will repost.

Yours,

Bowd

Subject	Re: Peanuts & The Tracker
Entry	12/05/2024 08:36 PM
Member	bowd57

The peanut shell from the last post was of course a tracker-defined critical intervention, and the bad nut an accident or blatantly illegal moving violation.

I'm a bit hesitant as to how to proceed from here, because I don't want to talk anyone into anything, instead I want people to engage with the tracker and form their own opinions and share them to inform mine. But I've never been good at subtlety, so I'm just going to spit things out. Please keep my intent in mind, though!

Look, the tracker is weak, but the requirements so high and the effects are so strong that that doesn't really matter. What we're trying to do is to guess when Tesla might have significant robotaxi operations. That depends on where they are, where they need to get to, and how rapidly they're progressing. The tracker can't help us with the middle term, but can with the other two. As we all know, the general form for this kind of thing, assuming you want to use historical growth rates to forecast future, given start value, end value, start date, end date, and target value is:

$\log(\text{target}/\text{end}, (\text{end}/\text{start})^{(1/\text{years})})$

I just put that in because I'm slow and it took me a while to figure out and now I have a place to find it if I forget.

I don't think you need the tracker to see that FSD has a long way to go. But the tracker gives us actual numbers, and as a skeptic I appreciate that the numbers come from Tesla superfans. The numbers are especially helpful when trying to guess at the growth rate. I'm not arguing that any number is better than no number. I'm asking for people to engage with the tracker to figure out how far off these numbers could be. Because my base case remains that Tesla needs to 1000x and is improving (charitably) at 40%/year, and thus is 20 years away. Those numbers could be way off! Let's try 64x improvement and 100%/year, that gets us to target in 6 years, which I think is the very edge of commercial relevance.

Does anyone have different numbers? Where did they come from?

It's the slow progress (well, slow relative to requirements! Doubling in 1/2 years is usually considered fast!) that dominates here. I don't find this surprising, because I think they still have a ways to go before having a truly first class ADAS.

I'm sure everyone is as bored with this conversation as I am. I still believe that if you're not engaged with the tracker (NB, WMC, I count calling it GIGO a dismissal, not an engagement) you're not engaged with the debate, but seem to be alone in that belief.

Yours,

Bowd

Subject	Trump Factor
Entry	12/05/2024 08:53 PM
Member	bowd57

Hi, all –

Forgot about this.

Before going on, no matter how tiring my droning on about the tracker is, it's not as tiring as, "Impossible without LIDAR!" "Why? Humans can drive with just two eyes!". I'm at least trying to inject some numbers into the conversation.

The Trump presidency, unless you think he's completely corrupt and that Musk has something on him, is prima facie bad for the company's legacy business lines. The pitch seems to be that it creates a better regulatory environment for FSD. I have a hard time understanding this, because there really is no Federal regulation of autonomous vehicles, so ... Am I missing something here?

Yours,

Bowd

Subject	Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 10:34 PM
Member	natey1015

If any VIC members decide to use this new FSD version when it's snowing outside with tough visibility, please let the VIC community know how it goes. Thank you!

Subject	Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 10:46 PM
Member	T0YPAJ182

"If any VIC members decide to use this new FSD version when it's snowing outside with tough visibility, please let the VIC community know how it goes. Thank you!"

Old short thesis: FSD is garbage that is nowhere close to Level 4 (appeared true a couple years ago)

Current short thesis: Let us know how FSD v13.2 does in the snow

Is this a bullish comment?

Subject	Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/05/2024 11:45 PM
Member	natey1015

I'm not bullish or bearish (have no position currently)--and completely open minded. I'd honestly like to know how the technology does in conditions when visibility isn't good since there is no lidar or radar used. Now we're into winter--hence the question about snow, but also interested in any other difficult driving conditions including rain, etc. That's it. No dog in this "fight." Thanks.

Subject	Re: Re: Peanuts & The Tracker
Entry	12/06/2024 12:16 AM
Member	RugPull1995

Bowd,

I've genuinely enjoyed reading your comments on TSLA as well as your engagements on other parts of VIC, so please take the below as a kind effort to convince you to end your short on TSLA. I was short TSLA in massive size for the first part of this year (think 10 digits big), and 9 months later, I genuinely cannot think of a worse short. I wouldn't touch this thing on the short side with a ten foot pole.

I'll address a few points shortly, but first, let's assess the kind of market we are currently in - I'll leave that at that since I assume you get what I mean. If wall street genuinely comes to believe over the NTM that TSLA has solved autonomy or is on a straight line path to solving autonomy on any remotely reasonable time frame (e.g., even five years), the stock will double, maybe triple, I don't know. You can argue a million different ways about how that valuation doesn't make sense at 2T or 3T market cap - it just is what it is. You know it, I know it - let's not kid ourselves on what happens if that narrative takes hold. They'll get full credit for the autonomous opportunity, and the call option value on Optimus goes up. Is what it is in this market. Such is life.

So that is the upside convexity if in fact you are wrong about TSLA's pace of progress on FSD (which we will get to in as second). But I guess more importantly, even if you are right, I have to point out, for a short to work, holders have to sell the stock. So what makes them sell the stock? What makes the world come to the realization that TSLA cannot solve the tech in some reasonable time frame? This is the problem - the FSD mega bull thesis isn't disprovable. Sure, they could take longer than people think, but I really don't think that matters - Elon tells a great story, and as long as people think the progress is up and to the right and increasing at an exponential (using that term loosely), people won't sell. Seriously, I was at the robotaxi launch event, I heard the retail investor in line talking about TSLA - they are absolutely nuts. More nuts than you can imagine. Think about what makes them sell. And wall street? Cathie Wood isn't dumping this thing because FSD will take another 36mo longer than people thought.

Second, you appear to have moved the goal post from "TSLA cannot solve the tech" to "TSLA cannot solve the tech **in a reasonable time frame**". I have to push back hard against that. I have an FSD DCF model where I create ludicrous, non-sensical, garbage in garbage out assumptions on miles driven, autonomous penetration, prices, revenue, the trucking opportunity, duopoly with Waymo share, blah blah blah, you get it - the whole kit and kaboodle. That isn't the point. The point is, when I did it, I realized, the opportunity is so damn big, that in basically any DCF, the terminal value in year 2040 or 2050 is literally all that matters. I encourage you to do a similar exercise. All that matters is if they get there, not when they get there. Time frame is mathematically given the size almost irrelevant.

Third, on your skepticism on the tech. You should really ask yourself, how sure are you? I mean really, are you sure they cannot get the tech to be safer than a human? TSLA IR is going around saying the tech is already at 10K miles per intervention on the latest version of FSD v13 they are testing in their fleet. Are you saying they are brazenly lying? I guess you are right? So let's make that clear - you are saying the IR department are borderline fraudsters. If that isn't the case, and your point is that they cannot go from 10K to 600K miles per critical interventions (which is human safety), I think that is just incredibly weak given the progress they are claiming to have made in literally 12 months since v12 launched. I should also add, as I said before, I went to the robotaxi event in LA, and while I was there, I cornered like 7 TSLA engineers (there were tons of them there) who work on FSD and had margaritas with them (I was quite drunk by the end). Nobody said anything they shouldn't have of course, but it was clear that the engineers themselves genuinely, truly, faithfully, like really really believe they are seeing massive progress in the technology with the new E2E deep learning approach, and boy oh boy they seemed confident it would continue. Are they lying? Sundar Pichai today at the NYT deal book conference said that "TSLA is making incredible progress" and that it is basically TSLA and Waymo at the top in the autonomy race. Is he an idiot? All that to say, you should really really ask yourself - how confident are you in your belief that they aren't going to get there?

I hope this was helpful. And hope to engage on some other idea in the future with you that is less frustrating and emotional than TSLA.

Yours,

Rugpull1995

Subject
Entry
Member

Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
12/06/2024 12:30 AM
RugPull1995

The technology fundamentally is "megapixels in, controls out". This is basically drive GPT. Just like Chat GPT was trained on an enormous corpus of human content and spits out human words, chat GPT is trained on an enormous corpus of "good" drivers (who they select for) where massive amounts of video data from human drivers goes into the system, and the system learns "when I see these kinds of pixels, I should do this". So for rain, snow, basically anything, the conceptual answer is that if a human can do it using eyes and neural nets, so can the FSD neural nets. They just feed tons and tons of videos of humans driving in rain and the system learns. Same with snow. The reason why it becomes safer is that it is estimated that something like 95% of accidents are the result of human error, not the vehicle malfunctioning. So the system learns good actions from humans, but doesn't text and drive, doesn't get tired or drunk, doesn't speed, etc. At least that is the theory.

On camera occlusions from snow, I have asked the company this Q before and they are pretty straight up about it. They basically said what would you do as a human if you couldn't see a single thing out of your windshield? You would slow down massively and in a worst case, try to stop driving somewhere. That is what FSD would do. This isn't meant to be some panacea that covers 100% of all possible edge cases that humans cannot do - the goal is to be significantly better than a human and that is the right standard. Beyond that, I guess there are like 8 cameras on TSLA's, so the chances of them all being occluded are pretty low, and TSLA is also working on camera heating tech that melts snow near the camera etc. Not perfect, but there are some work arounds.

For what it is worth, this is a problem for Waymo too. People think that using LIDAR is somehow an incredible solution for when snow covers all your cameras - it isn't. Waymo's system is designed to fuse all three sensor inputs into its perception system which feeds into its planning system. If the cameras are all occluded, the tech won't work as it is designed to take inputs from all three sensor types. The reason Waymo uses three sensor modalities is because they think it is the only way to make the car as safe as possible and will result in the cars always being the safest. We will see as that is the open debate - how much does the hardware really matter vs. the software (which is the AI brain)? If the brain becomes so good does the incremental value of the additional sensor modalities diminish? We will see and know the answer in 5 years or so.

Subject	Rugull
Entry	12/06/2024 02:46 AM
Member	bowd57

Hi, Rugpull --

Thanks. I'm personally grateful for your taking the time to share your very, ah, thoughtful thoughts, with which I'm in substantial agreement. I'm going to respond with some clarifications and points of possible disagreement, not line by line, but more by topic.

1: The stock. I couldn't agree more. All along I've been (unsuccessfully) trying to make a distinction between a cult stock and a meme stock, and claiming that Tesla was just too big to be a meme stock. That all looks pretty stretched right now. To my very meagre credit, I closed the idea out on VIC after some gentle prodding from Ray Palmer. But I should have closed the idea & my position in the Spring of 2023. I was lazy and complacent; "Oh, look," I thought to myself, "Tesla's getting a second wind from the AI bubble, I wonder how long that will last?" Whoops. I'm personally having a hard time covering at what I hope is peak meme, but that is a *personal* problem, and I already know what everyone is going to tell me. A short term problem I have is that I think the most recent 50% or so move due to the Trump victory is completely unmotivated. But that's just what meme stocks do. The 20-30% rise after the S&P inclusion was also fundamentally unmotivated, that didn't keep the stock from going up x00% from there.

2: The tech. This might get a bit long. I'm going to try to briefly tackle goal post moving, valuation, and odds of success. Some of these subjects are tied into each other.

First, let me address the narrow question of goal post moving. Note that I'm doing this in attempt to defend myself against charges of goal post moving! I'm thinking out loud about whether that's what's really going on.

> you appear to have moved the goal post from "TSLA cannot solve the tech" to "TSLA cannot solve the tech **in a reasonable time frame**".

At some point in my 9MM posts to myself, I said something like, "I've been involved in projects that were never going to ship, and from the outside, Tesla's FSD development seems exactly like a project that's never going to ship." Which, yeah. The remark was I think accurate, but insufficiently caveated. A fuller expression would have been, "With that team, that approach, that leadership, they're never going to ship." Obviously, approach, team and leadership are all subject to change. We've seen a change in approach that brought about dramatic improvements in ride quality and substantial improvements in, umm, need for constant management of the driving task. But WRT to autonomy, we're rapidly converging back to, "Never going to ship." So, listen with love and charity. There are people out there who say, "Impossible without LIDAR." Some of them are idiots. The ones who aren't mean, "challenging to achieve in a commercially relevant timeframe without LIDAR." The six year # I threw out was an arithmetic exercise/math test, not a forecast! The point was, "If we stress these inputs until I start getting obstreperous, where does that leave us?" Basically, I'm all kinds of crazy, but not crazy enough to say absolute things like "never", "commercially relevant time frame" should be assumed to be lurking in there somewhere..

The mention of commercially relevant time frame segues us into some thoughts/questions I had about valuation:

> the opportunity is so damn big, that in basically any DCF, the terminal value in year 2040 or 2050 is literally all that matters. I encourage you to do a similar exercise. All that matters is if they get there, not when they get there. Time frame is mathematically given the size almost irrelevant.

I'm not saying I disagree with you, but I can tell you who does: Google. And Toyota. And umm Baidu. And pretty much everybody but Woods and Musk. If your research has led you to that conclusion, I think you should get long here, because Musk is one of the very few who *get it*, who understand the opportunity and are looking to maximize it.

Probably not the time or place to get into the details, but I'm curious as to what you came up with. Like, margins & market structure just don't matter? I'm

probably missing something. Maybe you're assuming that Tesla will at least be a credible #2, ala Uber & Lyft?

I haven't spent that much time thinking about these questions, because what's really going on here is that, just like EVs, the market thinks this going to be HUGE, and Tesla is going to DOMINATE it.

Odds of success – I think this is interesting enough to promote to its own sub-topic.

3: Odds of success

If this were a fundamentally based stock, then I just wouldn't worry about this, because I think the tracker is close enough to bail you out from going down with the fundamental ship. Right? Nobody has a crystal ball! Tesla needs to see a real acceleration in progress for the FSD program to come anywhere close to meeting market expectations (which might not be yours), but if you can see that coming then, it's just, "Whoops, sorry gang, looks like I was wrong here," and then you cover.

I'm not sure what to make of – but am not doubting – your interactions with IR and Tesla engineers. I don't know what to say. Musk started talking about 5-10x improvements *right now*, I think in ... April? That was for version 12.4, which wound up never shipping. 12.5 has huge problems with running red lights and turning itself off due to glare. I'm too lazy to look to see if I'm parsing the remarks correctly, but I think Tesla has claimed that we've already *had* 5-10x improvements, which aren't visible to anyone. (These claims have led the faithful into an Easter egg hunt for what they might possibly be talking about.) This is a very personal opinion, but – I think something is badly broken over there, but am not close enough to guess as to what.

Rugpull, I'm running out of steam. I want to thank you again, and hope I haven't bored you to much if you've even made it this far. I hope this note won't be interpreted as an attempt to argue or defend. But out the door, and a pro pos of interactions w. IR:

> TSLA IR is going around saying the tech is already at 10K miles per intervention on the latest version of FSD v13 they are testing in their fleet. Are you saying they are brazenly lying?

Did you get them to define intervention? Remember, this is the company that claims 1 accident/7 million miles driven, but defines "accident" as "air bag deployment", and that boasts about 99.99% charging station uptime, but defines "up" as >=50% of the chargers working. Again, and it's too late for me to look up the details, I believe that they've made similar claims about versions that have already been released. Nobody knows what they're talking about. The cars still run red lights and turn themselves off.

And now I'm just going to say, thanks again, and apologies for droning on so long.

Your,

Bowd

Subject	Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/06/2024 09:01 AM
Member	natey1015

RugPull1995--thank you for a very thoughtful, insightful and helpful response--happy holidays!

Subject	Re: Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/06/2024 10:18 AM
Member	Weighing Machine Capital

RugPull,

Thank you for your depth of analysis. Autonomy is the hardest real world AI problem to solve (by far given the safety critical nature). I don't think there will be multiple autonomy players when all is said and done.

An analogy:

I am Google Pixel guy. I made the (in hindsight incorrect) bet a decade ago that iPhone users would eventually migrate to the "better tech" of Google/Android. I was wrong.

But, my experience using Android as my ecosystem for digital life is generally equivalent to Apple's iOS ecosystem.

HOWEVER, no one is going to want to have the second or third best autonomy platform to keep their families safe.

The horse that wins the autonomy race will reap most, if not all, the economics given the safety critical nature of autonomous mobility.

And the network effects will continue to reinforce the ultimate monopoly of the winner.

Tesla is most likely the winning horse. Try out version 13.2 when it goes wide release in a few weeks. It's nothing like prior versions.

CAN I TROUBLE YOU FOR AN UPVOTE FOR REACTIVATION SO I CAN CONTINUE TO LEARN FROM YOU GUYS?

I would greatly appreciate it.

Happy Holidays to All,

-WMC

Subject	Re: Re: Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/06/2024 10:39 AM
Member	finn520

WMC,

"The horse that wins the autonomy race will reap most, if not all, the economics given the safety critical nature of autonomous mobility."

Are you sure this is true? Say TSLA reaches what a layman would consider fully safe autonomy in several years, you click a button and reliably get to the destination, I believe that is considered Level 5. Then a few years later, a bunch of others follow the blazed trail and get there as well. I think at that point, maybe safety just stops being an issue because the technology is so good and accidents are down 95%+ from current rates.

An analogy would be something like steam engines in boats/trains (this is conceptual, I don't know the specifics) . When those first came along, they exploded all the time and lots of people died. You really wanted the best one because there was a huge safety difference. But once the technology problem of "not blowing up" was solved, I'm guessing there were a number of companies that sold them. Or another analogy - my understanding is that most commercial airplanes are on autopilot most of the time, but I don't think the owner of that technology has some crazy monopoly.

While we aren't there yet, it's not hard for me to imagine a world where once you get Level 5 autonomy, the technology gets so good, so fast, that safety stops really being an issue. Once you get to that point, why do network effects matter?

Subject	Re: Re: Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry	12/06/2024 10:50 AM
Member	Weighing Machine Capital

Finn,

Great points and question.

I think the network effects that the leading horse will accrue will expand the gap vs the other horses as take-rate increases. The flywheel of more cars on FSD > More and diverse data > solving edge cases > better ecosystem > more users.

It's not only about the pure software. It is an ecosystem between the car/sensors/on-board computer/cloud compute, etc.

And what is also not properly understood is Tesla's manufacturing advantage flywheel:

Tesla Manufacturing Prowess and Ecosystem:

Their car factories are designed to self-learn and continually optimize the manufacturing process. All their machinery is connected to a "Tesla Brain" that continuously performs A/B testing to simulate experiments with minute changes to their processes. This creates a flywheel effect, resulting in an increasingly efficient manufacturing process. No other manufacturer in the world comes close to this. This is truly Tesla's number one barrier to entry. No one can manufacture electric vehicles as efficiently and autonomously as Tesla can today, and with their self-learning architecture, their lead over other manufacturers continues to grow. This ecosystem in car production will be the same one Tesla utilizes for their humanoid robot manufacturing, and in my strong opinion, no one else will be able to replicate their self-learning manufacturing optimization flywheel. The Total Addressable Markets (TAMs) for both autonomous cars and humanoid robots will be in the tens of trillions of dollars. In my strong opinion, Tesla is likely to be worth more than \$10 trillion in the next 5-7 years.

Subject Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/06/2024 10:55 AM
Member cuyler1903

WMC - Both Google Pixel owners who also own Zillow?! I knew I liked the way you think!

Cuyler

Subject Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/06/2024 12:23 PM
Member heffer504

Sorry to seem like I'm picking a fight here-- I'm not.

But do you remember when Musk spoke about the Alien Dreadnaught factory based on first principles and then was making cars under a tent for a few years?

Also, have you heard of BYD? Aren't they (and other Chinese manufacturers) demonstrably more efficient than Tesla?

Is your "strong opinion" that "no one else will be able to replicate their self-learning manufacturing optimization flywheel" based on an exhaustive study of global manufacturing processes?

Free advice-- If you are going to speak in meaningless and indefensible blue sky language, then try the sell-side, venture capital, or maybe ARK. I'm not sure Value Investors Club is the right venue for you...

More broadly, I think it's interesting that everyone who is discussing the bearish side of the argument here isn't actually short the stock (myself included). If anything, this is potentially bearish-- short sellers have moved on but can always get back involved!

Subject Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/06/2024 02:35 PM
Member miser861

Maybe the real alien dreadnaught factory was the friends we made along the way

Subject What?
Entry 12/06/2024 02:46 PM
Member bowd57

Hi, WMC --

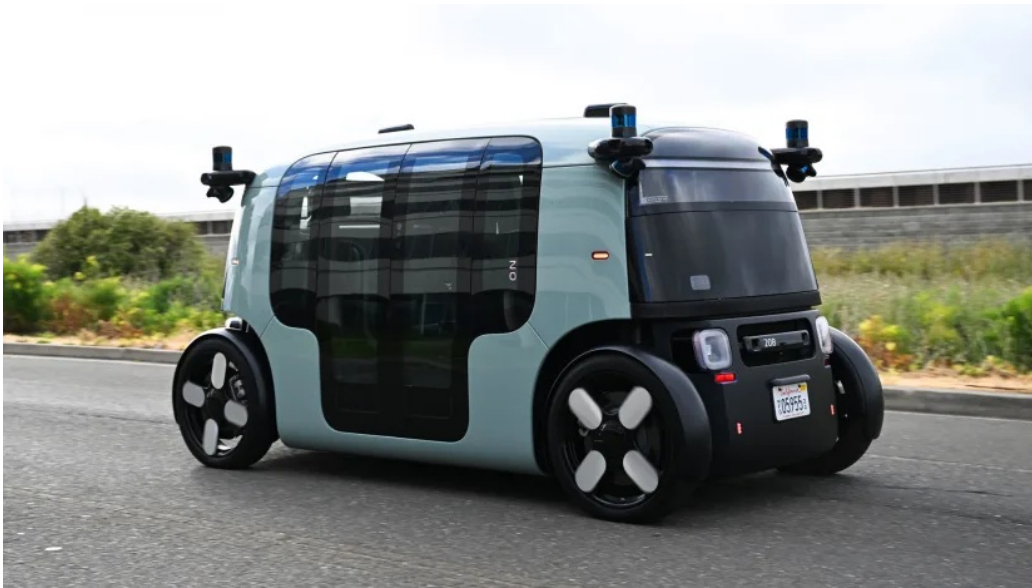
> I think the network effects that the leading horse will accrue will expand the gap vs the other horses as take-rate increases.

Our views of reality are so different that I'm not sure we can have a productive conversation. Or wait -- you can't be thinking of robotaxis here. Or maybe I'm conflating your views with Finn's question:

Finn:> Say TSLA reaches what a layman would consider fully safe autonomy in several years, you click a button and reliably get to the destination,

Finn, this has already happened. Again, maybe you're thinking of something else. I'm talking about robotaxis. That's a service where you charge consumers for taking them from point A to point B. The novelty is that there is no driver needed, which is expected to deliver better economics when scaled.

Multiple companies are operating driverless taxi services in multiple countries on multiple continents. Many others are talking about entering the market. Tesla is one of them, but there are many others. For example, there's Zoox. Here's a picture of one of their vehicles:



I bring up them up because they're owned by Amazon, a company you may have heard of, which is thought to a formidable competitor. People who seem to have industry knowledge say they're serious and legit, and that we shouldn't be deceived by their low profile to date. This is unverified by me, I'm just passing this along.

The long pitch I'm familiar with for Tesla in autonomy can be briefly stated: Tesla doesn't have robotaxis right now, but they will soon, and when that happens, they'll have an insuperable advantage stemming from radically lower capital costs due to requiring less compute and fewer/cheaper sensors. Furthermore, the ability to use their existing consumer fleet will grant them insta-scale, thus strangling any potential competitors in the crib. If you guys are thinking of a different pitch, let me know.

Rugpull, the above pitch is why "commercially relevant time-scale" is important. Given enough time, hardware costs will come down or competitors figure out how to hit their safety numbers with less hardware. The "insta-scale" part loses its force if Waymo or others are already well established in the top N markets. You may be saying that the market will be so large that anyone who's a player will go through the roof, I'm not sure. WRT to industry valuations as a whole, Tesla is trading at multiple times the entire rest of the publicly traded comp set, just like with auto back in the day. Note that a lot of big participants are private or locked inside large companies. Waymo recently raised \$5B; I'd be interested in any scuttlebutt about the valuation.

Anyway, a lot of the recent conversation about flywheels, etc., doesn't make sense to me given my understanding of what's going on.

This message is getting long so I'm going to arbitrarily truncate it here and make a couple of more points in a follow up.

Yours,

Bowd

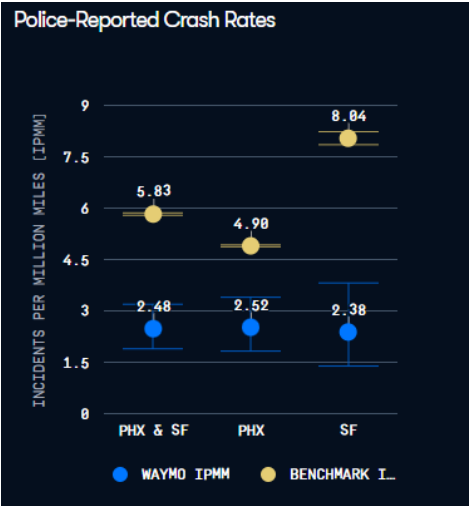
Subject
Entry
Member

A Few More Points/Questions
12/06/2024 02:57 PM
bowd57

Hi, all –

1: That Tesla is talking about 5-10x safety improvements is at some level a tacit admission that that level of improvement is needed. If they can roll out a couple of 10x releases, then, yeah, then we're really talking, and my hats off to them. I think they'd still want another 10x, but wouldn't see why they couldn't scrape that together over the next couple of years. The problem is, I need to double check, I think they've claimed that such improvements have already occurred, it's just that nobody can see them. At some point it gets Canadian-girlfriendish: "Oh, you wouldn't know her, she's in the next release."

2: Finn & WMC, both of you have expressed an interest in safety. Here's a chart from Waymo showing where they're at lately:



You can find much more here:

<https://waymo.com/safety/impact/>

To get to those levels, you have to pretty much eliminate at-fault crashes and take a big chunk out of other people smashing into you.

3: WMC:> All their machinery is connected to a "Tesla Brain" that continuously performs A/B testing to simulate experiments with minute changes to their processes.

Is this disclosed or discussed anywhere? Did I miss a conference call or presentation? It sounds interesting, I like to learn more about this Brain.

Yours,

Bowd

Subject
Entry
Member
<https://youtu.be/OJ8CXsuHBzk?si=1g4zGVjKzB0EoHYX>

Re: A Few More Points/Questions
12/06/2024 03:29 PM
miser861

Subject
Entry
Member

Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
12/10/2024 12:54 PM
miser861

One of my cars is a Tesla and I can assure you FSD is not ready for a commercial unsupervised taxi service. Is it really good for supervised retail driving? Yes. Is it a reason to buy the car? Yes. I don't know where Tesla gets the once in 10,000 miles disengagement stat because it's way more than that. Maybe on a closed course? As just one example, my neighbor has these blue landscape lights and every time I drive by their house FSD disengages and the windshield wipers turn on. A paying customer in an unsupervised car would probably think that's way less funny than I do.

Subject
Entry
12/10/2024 02:14 PM

Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
12/10/2024 02:14 PM

Member

Weighing Machine Capital

You'll see what I'm talking about within a few weeks when v13 goes wide release.

Subject

Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED

Entry

12/10/2024 02:54 PM

Member

amr504

"You'll see what I'm talking about within a few weeks when v13 goes wide release."

The evergreen TSLA FSD comment since 2017... just replace the number behind the v.

Subject

Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED

Entry

12/10/2024 03:24 PM

Member

bowd57

Hi, Miser --

One of the amusing things about following the FSD story is the number of people who incapable of understanding that others might have different experiences. I don't know how many times I've seen versions of this conversation:

"Wow, that's weird, works great for me. There must be something wrong with your car. Did you recalibrate your cameras?"

"Yup."

"Clean the windshield?"

"Yup."

"Take it to the shop?"

"Yup."

"Well, you should take it back in. I don't know what's wrong with your car, it works great for me."

Yours,

Bowd

Subject

no position: but seems like the market is acting rationally

Entry

12/10/2024 04:11 PM

Member

zach721

Musk invested \$260 million, and the market is just pricing in a quid pro quo/reciprocal benefit to Musk/Tesla. It seems like the market thinks it could be worth \$500+ billion, and it could be right. Tariffs, regulatory oversight, subsidies for EVs, and the ability to impact competitors—I mean, it's everything for TSLA, right?

**CNN** Politics

SCOTUS

Congress

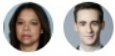
Facts First

2024 Elections

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Musk spent more than a quarter-billion dollars to elect Trump, including funding a mysterious super PAC, new filings show



By Fredreka Schouten, David Wright and Alex Leeds Matthews, CNN

5 minute read · Updated 9:33 AM EST, Fri December 6, 2024

**Subject****Entry****Member**

Re: no position: but seems like the market is acting rationally

12/10/2024 04:26 PM

bowd57

Hi, Zach --

Tough to figure. A lot of this depends on how corrupt -- I'm open to a gentler word! -- you think Trump is, and how fickle, again open to another word.

Trump's on record that he wants to cut EV & green power subsidies, which strikes at the heart of Tesla's legacy auto & energy storage businesses. This has recently been affirmed by people who should know, I'm thinking in particular of newly elected auto dealer senator here. Could he say, "Cut, except for Tesla?" Sure, I guess that could happen.

There may be a payoff for the autonomy business. In theory, this doesn't need anything, because, 25k miles to necessary intervention and going up! But a "look the other way" regulatory posture could certainly help to umm cover things up should deployment outpace progress, if you know what I mean.

As I've pointed out before, Musk probably thinks there's some tangible benefit for him. Maybe that will happen, maybe not; let's ask Chris Christie! But that doesn't mean that the tangible benefit winds up at Tesla. Space X and even the Boring Company seem like better fits for Trump's public postures.

One final thing about robotaxis -- Open up a county level presidential electoral map. All those blue dots lost in the sea of red are the places it's most profitable to run a taxi service. If Tesla wants to run a robotaxi service in, say, San Francisco, how do they even brand the thing? As Tesla? As X Car? Musk better get to the lab and invent some egg-repellent paint!

Note that bad long <> good short.

Yours,

Bowd

Subject Re: no position: but seems like the market is acting rationally
Entry 12/10/2024 04:32 PM
Member miser861
 First thing Trump should do for Elon is force Grimes to take a bath

Subject Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/11/2024 02:08 PM
Member miser861

Sounds like it's significantly smoother and more assertive than v12. That's a huge issue. His wife says it's her first FSD drive with no disengagements. There's still way too many interventions to be Level 4. In another video he did yesterday there was a disengagement due to sun glare blinding the forward camera, that's clearly a hardware deficiency. It's a great product for consumers, but not robotaxi ready.

<https://www.youtube.com/watch?v=amNon8kigSw>

Subject Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED
Entry 12/11/2024 03:58 PM
Member natey1015

Thanks for sharing. That is pretty awesome. If you or anyone else sees videos of people driving V13.2 in tough weather conditions like snow, hail, rain, fog, etc. please share the link--really curious to see how the technology does then.

Subject FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves
Entry 12/18/2024 11:47 AM
Member Weighing Machine Capital

Transporting people and things around the planet autonomously will change our lives and the economy in such profound ways.

Experiencing FSD v. 13 will be the moment you realize Tesla won the race to autonomy.

Subject Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves
Entry 12/18/2024 12:06 PM
Member amr504

LOL... "strongly suggest"

This Christmas season, all I ask for is the confidence of a Tesla shareholder.

Subject Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves
Entry 12/18/2024 01:39 PM
Member bowd57

Hi, WMC --

> Tesla won the race to autonomy.

<https://waymo.com/safety/impact/>

Maybe you mean something non-standard by "race" or "autonomy" or "won".

Yours,

Bowd

Subject Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves

Entry 12/18/2024 02:19 PM

Member natey1015

I recently watched several V13.2.1 videos of people driving--one of which was in the snow. It looks like it's a massive improvement compared to version 12. That being said it's clear to me that if the cameras' vision is not 100% clear then level 5 autonomy goes out the window. Compare this to Waymo due to lidar where that is not the case and can remain fully autonomous in bad weather conditions where visibility is impaired.

So here's the way I look at TSLA's driverless technology as it stands. One can have a lot more confidence in clear weather conditions that it's near level 5. I guess that means if you live in a warm weather area with little rain or snow then it can be a real solution to free up one's time when driving assuming you're allowed to remove your hands from the steering wheel and eyes off the road--which currently you cannot do.

TSLA wants you to pay \$99/month for FSD. That's a steal if it can actually be an autonomous solution during good weather times. I could see a world where I have my TSLA for good weather (using it say 75% of the time) and supplement it with me driving during bad weather times or using Waymo.

Now what happens for the times when I know I'm going to have a few drinks and don't want to drive (assuming the weather is good)? Until the law changes, I can't use my TSLA FSD without risking a DUI. So Waymo is still the best solution here.

Personally, I think the holy grail is Waymo offering its technology to OEMs so consumers can own that technology in their cars and/or rent it through the app. We know there is a meaningful additional cost to having lidar, but that cost has been and continues to come down a lot. I'd gladly pay an extra ~\$200/month more for "FSD Waymo with lidar" knowing that my car can be a true chauffeur anytime I want it to be. I look at it as essentially life insurance. It solves the drinking and driving issue. It would allow it to take my kids to/from places if my wife or I couldn't drive them. And if you could rent out your car by putting it into the Waymo system to make money for you when you're not using it then consumers would be willing to pay a lot more for the technology since the net cost would be much less or possibly nothing (or even a net positive).

Musk has been vehemently against using lidar because of the cost. However, is it out of the realm of possibilities for him to switch his mind in the future if the cost of lidar comes down enough?

As a consumer who owns a Tesla, I have zero interest in FSD until it can truly be level 5. And until I see proof otherwise, it can't be 100% level 5 without lidar. If I still have to sit in the driver seat with my hands close to the wheel and eyes mostly on the road, what's the point of that? Maybe Musk will get the law changed. Still I wouldn't trust it without lidar, but that's just me.

Subject Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves

Entry 12/18/2024 03:27 PM

Member miser861

<https://youtu.be/ar2rpDQG47M?si=DMoM4Y9DyJeDZ57Q>

Subject Re: Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves

Entry 12/18/2024 03:28 PM

Member Weighing Machine Capital

Waymo only works in geo-fenced & pre-mapped areas. What happens when the driving environment changes due to road closures, construction, fallen stop sign, etc?

Waymo doesn't SCALE. Tesla's FSD is usable no matter where you are on the planet. And the car is manufactured at a positive Gross Margin. Go buy a Waymo-equipped car. You can't but they cost >\$150k.

Autonomy is not a hardware problem at this point; it's a software problem and scaling laws of more compute/larger model/more optimized model/more training data will get to significantly better than human.

A good analogy is the exponentially improved Chat GPT o1 model vs. 4o. o1 is built with a significantly smaller training run/model size. Then how is it better at reasoning?? AI research at OpenAI has improved the optimization of the model. Tesla is doing the exact same thing with the best machine/deep learning engineers in the world.

Say what you want about Elon, but no one on the planet can marshal human resources to accomplish a near impossible goal (rockets/FSD) like him.

Subject Re: Re: Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves

Entry 12/18/2024 03:46 PM

Member natty813

So this has been a great rec. Hats off to Weighing Machine capital for the write-up and engagement.

I read the Isaacson bio and have followed Musk closely. Lost money shorting TSLA many years ago in a harrowing experience. Modern day Howard Hughes-

Also, I know it's 2024 and things are unhinged. But at \$1.5 trillion, under some normal circumstances, wouldn't we consider FSD and all sorts of other goodies discounted in the price?

Again - great pitch

Natty

Subject Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves

Entry 12/18/2024 04:59 PM

Member bowd57

Hi, WMC --

> What happens when the driving environment changes due to road closures, construction, fallen stop sign, etc?

That's a very good question! Why don't we ask one of the 150,000 people who take a Waymo trip every week?

> scaling laws of more compute/larger model/more optimized model/more training data will get to significantly better than human.

Have you looked at these scaling laws? How much does compute need to increase to halve the error rate? Model size? Data size? Scaling laws are not pretty!

> Say what you want about Elon

I have no desire to say anything about Musk.

WMC, don't take this the wrong way. I've read your other work and think it's pretty good. But there's something about FSD that just breaks peoples brains. Tesla has not demonstrated anywhere, at any time, sufficient reliability for autonomy. That might change! Maybe v13.2 will be the one! We could discuss what the odds are. Maybe if/when that changes, Tesla will sweep the field due to lower capital costs. Could be! But right now, there are multiple companies running commercial driverless taxi services in multiple cities in multiple countries on multiple continents, and Tesla isn't one of them.

Yours,

Bowd

Subject Re: Re: Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves

Entry 12/18/2024 05:58 PM

Member natey1015

Wrong--Waymo is currently in geo-fenced areas to appease regulators and maximize fleet efficiency/ROI. As far as what happens when the driving environment changes due to something such as a "fallen stop sign"...https://x.com/dmitri_dolgov/status/1868778679868047545?s=42

To say Waymo doesn't scale is absurd. Look at how much the cost to produce a Waymo car has come down recently...<https://waymo.com/blog/2024/08/meet-the-6th-generation-waymo-driver>

And your statement assumes you can predict the future perfectly. That would be akin to me saying Tesla will NEVER produce a level 5 autonomous vehicle because it currently doesn't have the technology that can do it. I would never bet against Musk and his team of engineers.

All I'm saying is that there is zero proof Tesla's FSD technology (despite how much better V13.2.1 is) can get to level 5 autonomy under ALL kinds of driving conditions (particularly bad weather where the cameras' visibility is impaired). It does seem to be darn close in good weather conditions. And if I can't be chauffeured in it to better use my time reading, etc. then what good does it do me?

Subject My son

Entry 12/18/2024 07:25 PM

Member heffer504

I was discussing this with him (he's 16) and he made a nice comment:

Maybe all the rich guys on that site think self driving is such a big deal since they don't know that there is already a technology that transports you at a low cost and lets you watch YouTube at the same time. It's called a bus.

Subject Re: Re: Re: Re: Re: Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED

Entry 12/23/2024 02:28 PM

Member heffer504

T0YPAJ182, this is my first quarterly check-in to ensure that you have not been in an airplane (other than a Joby S4) and are not flying anywhere for the holidays. Please confirm this as per our wager. I can confirm I have not been in a fully autonomous Tesla (largely because there is no such thing!)

Regards, Heffer

Subject Entry Member Hi, Heffer --	Re: Just experienced FSD version 13.2 --> AUTONOMY SOLVED 12/23/2024 09:40 PM bowd57
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> I can confirm I have not been in a fully autonomous Tesla

You should be safe for a few more months; 13.2 doesn't seem to be The One.

Yours,

Bowd

Subject Entry Member Yeah but who's gonna set you on fire in an AV?	Re: My son 12/30/2024 12:53 PM miser861
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Subject Entry Member My 500 mile assessment of FSD v13 is that it very obviously still isn't Level 4. It's a linear improvement over v12. It's a very good - and improving - ADAS, but to say they've "solved autonomy" is clearly delusional at this point. I recommend you ride in a Waymo to see what God tier autonomy is like, it will give you a whole new benchmark.	Full Self Deluding v13 12/30/2024 01:12 PM miser861
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Not gonna type out my full notes, but some flaws I've noticed:

-frequently cuts other drivers off

-hits curbs

-has what I would call a Mario Kart driving style

Subject Entry Member Anyone who's thinking about saying "the battery pack", do not start with me. You have to get burned alive somewhere, it might as well be in the comfort of your own car.	Re: Re: My son 12/30/2024 01:31 PM miser861
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Subject Entry Member Mario Kart driving style- can you quantify more? In college, my friends and i would try to chug a beer every lap (for those unfamiliar, there are three laps in a race, with each taking a minute or two).	Re: Full Self Deluding v13 12/30/2024 01:36 PM Ray Palmer
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Is FSD "mario kart" like driving like me before the first beer? Because I am **really** good at Mario Kart; I'd probably want to drive in it if so.

In contrast, if it drives like me at mario kart after three or four rounds of drinking a beer per lap..... well, I'm surprised we haven't heard more about the slaughter on our streets.

Subject Entry Member Before the first beer. It seems to over-prioritize maximizing its speed within the speed limit. So if that means doing tons of lane changes to avoid doing 68 in a 70 that's what it'll do. It has no chill. I wouldn't say it's a safety risk at all, but the lane changes definitely piss off other drivers and will eventually get you shot where I live.	Re: Re: Full Self Deluding v13 12/30/2024 02:23 PM miser861
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Subject Entry Member Before the first beer. It seems to over-prioritize maximizing its speed within the speed limit. So if that means doing tons of lane changes to avoid doing 68 in a 70 that's what it'll do. It has no chill. I wouldn't say it's a safety risk at all, but the lane changes definitely piss off other drivers and will eventually get you shot where I live.	Re: Re: Re: Full Self Deluding v13 12/30/2024 06:03 PM cuyler1903
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I've actually got an order in now for another MX LR - will be my 3rd one. The first one got rear ended, and in the stimmy bubble I still sold it for \$20k more than I paid. The 2nd one (a 2022), still has HW3 and at 35k miles, I didn't want to drive it another year to the end of the warranty. So, waiting for the 2025 VIN to be issued (again) as I write.

Despite my loyalty, the good people at Tesla are testing my patience. I put in my order on 12/17 in order to take delivery by 12/31 and get some small incentives (3 months FSD + 3 months supercharging) if I took delivery by then. I'm trading my 2022 and getting \$50k for it despite it having 35k miles, so not bad. I paid \$80k for it and drove it for 3 years so \$10k/yr.

So, in the app, I get the trade value finalized, upload my insurance card, and then my 2025 VIN gets assigned a couple days before Christmas. I wire in my \$34k balance due, and think I'm getting my car by year end. Then, I look in the app a day later (after I sent my wire), and these jokers pulled my VIN from me. Probably because they couldn't get the car to the East Coast in time for year end, and decided to deliver it to somebody nearby instead. Now, Tesla is sitting here with (1) my order, (2) my trade confirmation and (3) my money, and I again have no VIN assigned or idea if/when the car will be delivered before my trade value expires. To add insult to injury, I get a text message from them saying that I can take delivery of a different car (with different specs I don't want) otherwise I'm also going to lose my incentives. Feels a lot like a bait and switch.

Regardless of all this, I'm looking forward to having HW4 and figure I'll subscribe to FSD on a monthly basis, as needed. I'm sure it's great, but there's no way it's at Waymo level.

If anyone has had a similar issue and a good solution, let me know.

Cuyler

Subject Re: Re: Re: Full Self Deluding v13
Entry 01/03/2025 01:57 PM
Member miser861
 The algorithm is gently nudging you to get a Cybertruck instead

Subject Re: Full Self Deluding v13
Entry 01/03/2025 01:59 PM
Member miser861
 Very clearly isn't ready for unsupervised driving

<https://x.com/SkepticElektrik/status/1872413955228119239>

Subject Read this post on X and the article at the end
Entry 01/08/2025 09:19 AM
Member Weighing Machine Capital
<https://x.com/pbeisel/status/1876657870533783645>

Subject Re: Re: Re: Full Self Deluding v13
Entry 01/10/2025 09:33 AM
Member cuyler1903

Update: after numerous emails from the delivery center to operations, Tesla did the right thing and re-assigned the promotion. Obviously only worth a few hundred bucks, but in principle was glad to see it. Only a few days in, but I do feel like the build quality is improving greatly, interior noise is minimal, car feels a lot more premium, and FSD seems to work much better. Ymmv.

Subject Re: Read this post on X and the article at the end
Entry 01/10/2025 03:48 PM
Member bowd57

Hi, WMC --

For those who didn't follow the link, the poster thinks that NVidia's chipsets are a snare, a delusion, a dead end, and that only the Tesla Way can get us True Autonomy due to Right Thought, Right Speech, Right Action.

Interestingly, a number of Chinese players -- too lazy to dig up a link -- who started with NVidia have brought hardware development inhouse.

I've wandered across seemingly informed commentators who think Tesla will be driverless around '27-'28 because everyone will. If they're right, it's hard to guess how things shake out. Maybe Zoox "wins" because Amazon can bundle it with Prime?

Yours,

Bowd

Subject Re: Re: Read this post on X and the article at the end
Entry 01/14/2025 06:00 PM
Member Weighing Machine Capital

@bowd57,

Can you share your "seemingly informed commentators"

Some of my go to sources for all things autonomy are:

@jamesdouma

@a_meta4

@TeslaTrip

@M_IsForMachine

They are, in my opinion, the best machine learning experts on X.

Subject Entry Member Hi, WMC --	Re: Read this post on X and the article at the end 01/15/2025 02:47 AM bowd57
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> Can you share your "seemingly informed commentators"

Most recently, redditior Recoil42. I've heard similar speculation/japery from Brad Templeton, and others who I'm forgetting.

Note that I'm not endorsing this! But it could happen. There are people other than Waymo testing in the US; Zoox and May Mobility come to mind. Mobileye is hoping to release Chauffeur around that timeframe. The Chinese are moving fast and there are a lot of them. Tesla might surprise me and get it together quickly. The industry could well wind in Cambrian explosion situation, leaving people scratching their heads and asking, "Where did all that come from?"

Yours,

Bowd

Subject Entry Member Brad Templeton is a strategy consultant for Waymo by the way.	Re: Re: Read this post on X and the article at the end 01/15/2025 11:27 AM Weighing Machine Capital
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Recoil42 doesn't seem to have any machine learning background.

Autonomy is a machine learning/AI problem to solve. I suggest you read what the leaders in the field of machine learning have to say and be wary of experts with biased agendas.

Subject Entry Member Hi, WMC --	Re: Read this post on X and the article at the end 01/15/2025 03:09 PM bowd57
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> be wary of experts with biased agendas.

Yes, that's excellent advice. You may want to scan through your list of experts and cull those who, for instance, were extremely bullish on Tesla & FSD *prior* to the introduction of "E2E", or those who are monetizing their views via Twitter or YouTube, and thus might be more properly considered "influencers" than "experts".

More substantively,

> Autonomy is a machine learning/AI problem to solve.

I want to stop you right there with an observation and a question.

1: The observation is, there's a lot more to it than that.

Let's stipulate that neural nets, for instance, are sufficient for AGI. Whoever demonstrates this can expect an extremely gratifying call from Sweden at some point over the next few decades, but let's just take it as a given. So, (A): Any problem can be "solved" (I hate that usage) by neural networks, (B): Personal vehicular autonomy is a problem, (C): Thus personal vehicular autonomy can be solved by neural nets.

Great. The problem with the solution is that you still need to *embody* it in some commercially feasible way on a commercially relevant timeframe. I'm reminded of the endless and pointless "Vision is only is impossible!" debates. The question isn't, "Is autonomy possible using cameras only?", but, "Is autonomy possible using 8 1.2 megapixel cameras from 2018?" The bull case here -- and, again, let me know if there's another bull case -- is:

A: Tesla will have an insuperable hardware cost advantage, and

B: Tapping into the customer fleet will give them sufficient insta-scale to strangle any potential competitors in the crib.

B has already taken a bit of a hit from doubts about whether autonomy can be achieved with HW3, ie, most of the fleet, and the announcement that company

plans to commence testing with safety drivers in a couple of locations in 2025, which doesn't sound like 1MM robotaxis waking up over night with a single software update. But perhaps insta-scale has just been delayed. The real problem is with A. They need to get things to work in a safety-critical real time environment with a particular set of sensors, compute, etc.

The syllogism I laid out above is just good logic; if you accept the premises you'll accept the conclusion. But here the argument seems to be, (A) something is theoretically possible, therefore, (B) Tesla will be able to do it quickly and cheaply. I don't think Aristotle would approve.

2: The question. This isn't a trap. It's something I've never understood. Somebody's missing something and maybe it's me. By way of background, I have experience in software development, and was quite the crackerjack C++ programmer back in the day, but I disclaim any knowledge of anything relating to AI – genetic algorithms, simulated annealing, neural networks, machine learning, the whole shebang.

Some Tesla fans point to scaling laws, generally derived from experience with LLMs, as evidence of the inevitability of Tesla achieving autonomy, but I've never understood why. I'm going to present what I think is the canonical chart of LLM scaling laws:

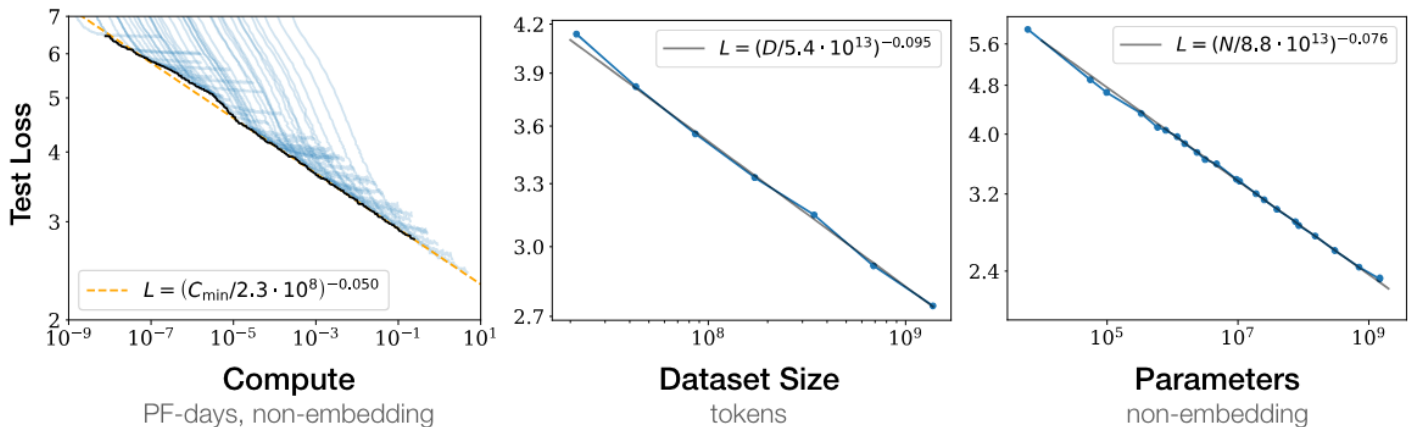


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

Maybe I don't understand the chart. I'd grossly oversimplify it as: There are three dimensions. For each one, you can cut the error rate in half by increasing the input by between 2 and many orders of magnitude. I'd imagine the results are multiplicative? So 2-to-many on each dimension would give you an 8x improvement? I don't get it. Like, the company announces that the latest release features a 3x increase in parameter count, and many say, "Yes! Robotaxi is here!" But when I look at the rightmost chart, it doesn't seem like 3x or 5x or 10x would really move the needle. What am I missing?

WMC, thanks as always for the conversation.

Yours,

Bowd

Subject Chinese AI
Entry 01/26/2025 09:28 PM
Member bowd57
 Hi, all --

I'm going to stick this here because it's as good a place as any and the jumping off point is about autonomous vehicles, which accounts for the vast bulk of Tesla's value these days.

A few days ago Redditor Recoil42 posted his take on the state of AI/autonomy in China. This has become an extremely hot subject due to the release of Deepseek, a Chinese open-source LLM that many think is the bee's knees. I'm presenting it because of its relevance to Tesla and intersection with other topics that are of interest to the club. And here it is!

https://www.reddit.com/r/singularity/comments/1i8tmh/okay_lets_talk_about_china_l2_l3_adas_in_2025_and/

And here are some notes.

1: I follow Recoil42, but I'm not fucking the guy or anything, he's not my hero or guru. As WMC points out, he's not an expert. But he is worth listening to. He caught my eye a couple of years ago with his piece on Chinese EVs which I can re-up if you can't find it. There wasn't much new there if you were paying attention, but it was a well put together summary that hammered the point home in an understandable way that also turned out to be very, very right.

2: This one is a survey of Chinese developments in, it's called different things, let's really piss the hard-core AV people by using "Level 2+". This more or less what Tesla's FSD is, that is, driver assistance that more or less works off-highway. Me and WMC were discussing, "Everybody will have autonomy in 3 years," this is very apropos of that conversation. There's once again not a lot of news if you're paying attention, but subsequent events and some of the claims he makes give the post relevance beyond just autonomy.

3: I'd sum up the claims with his section heading, "A Growing Expertise in General AI". I'll not summarize further, you can read it yourself, it's not long.

4: Point #3 and the visible success of Deepseek are obviously significant to what has been the biggest investment theme over the last couple of years.

5: The "Chinese" part opens a whole 'nother can of worms. This is a hard topic, because there's tons of -- I don't know how to say "Tankies" in Mandarin, but a bunch of China triumphalists out there, along with another whole bunch who just say, "Slave labor! Mind control! Social credit!" But ... Let me put this bluntly: What if the Sino-Tankies are right? Do you wind needing to invest in China, despite the awful awful political backdrop, because that's where the growth & innovation & advancement are?

Yours,

Bowd

Subject Too Dismissive
Entry 01/26/2025 10:14 PM
Member bowd57
 Hi, all --

Me:> again not a lot of news

This is unfair. Recoil42 does good work and has a grasp of the details that I didn't/don't, like, "Oh, you wanted some sources to back up those claims, what are you, some kind of pedant?"

Yours,

Bowd

Subject international
Entry 02/11/2025 09:29 PM
Member heffer504

OK, I couldn't resist shorting a little of this last week, mostly on the thesis that there will be ~ zero non-Americans who feel warmly towards Musk after his genocide-adjacent budget cutting priorities, and ~ half of Americans are also out of the market for a facism-mobile.

Does anyone disagree? Are there a bunch of international buyers somewhere who are thinking, "Man I really want to support Trump and Musk right now?"

Subject Is It Safe?
Entry 02/11/2025 09:59 PM
Member bowd57

Hi, all –

https://www.youtube.com/watch?v=xfgVz_Wan3w

Yours,

Bowd

Subject	Re: international
Entry	02/12/2025 01:07 AM
Member	churchill

I am not sure about people out there but Chrystia Freeland (Canada liberals' candidate) already floated the idea of targeting Musk in retaliation for the US tariffs, and I have heard similar proposals in Europe. To me the issue is that current valuation already discounts to a big extent that robotaxis will be a success soon, so I struggle to see much upside from here.

Subject	Sales imploding globally
Entry	03/06/2025 01:12 PM
Member	natty813

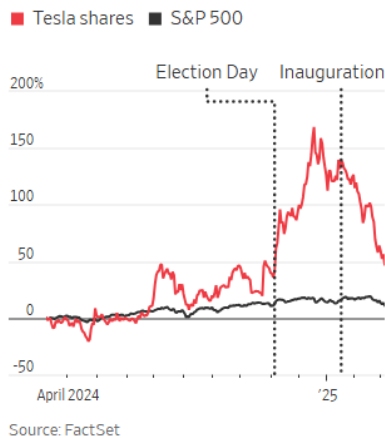
I love the work he is doing at DOGE, but Elon might want to consider putting the chainsaw down and returning to the office.

It is very clear that his politics are having a real impact on TSLA sales.

MAGA voters will not be dropping their F150s and GMC Sierras to buy CyberTrucks. But liberal and foreign buyers are going elsewhere and then there is China where they are getting murdered competitively -

Subject	Re: Sales imploding globally
Entry	03/07/2025 09:55 AM
Member	natty813

Change over the past year



Tesla, the nation's largest producer of electric vehicles by far, has long led the auto industry in consumer loyalty, and surveys show many owners have no intention of abandoning the brand now. Yet recent polling shows that the brand's appeal has been eroding, and sales data suggest it has begun taking a financial toll.

Back in 2022, before Musk waded into presidential politics, automotive consulting firm Strategic Vision found that 22% of car shoppers surveyed said they would "definitely consider" a Tesla for their next vehicle purchase. That put Tesla on par with other luxury car brands such as Mercedes-Benz and

BMW, said Alexander Edwards, the firm's president.

By last summer, that percentage had dropped to 7%—roughly in line with Lincoln and Dodge—and it has stayed there, a precipitous decline that is rare in the car business, Edwards said.

Subject	Re: Re: Sales imploding globally
Entry	03/07/2025 01:02 PM
Member	natty813

Just heard a new term:

Musk "Bud Lighted" Tesla--

Subject	Re: Sales imploding globally
Entry	03/07/2025 04:19 PM
Member	UCB1868

Ha. That makes no sense. You admit that Musk is hurting Tesla...but you want him to go back to Tesla? Do you think people are just going to forget? They'll be reminded every time they see a closed National Park or VA Hospital. You may "love" that a depraved South African is illegally firing Park Rangers and NOAA scientists that he never met, but most people I know are outraged. If Tesla's board cared about protecting the company, it would have disavowed Musk a long time ago (or at least after the first Nazi salute). It's way too late, buddy.

Subject Re: Re: Sales imploding globally
Entry 03/10/2025 10:51 AM
Member natty813

UCB - I'm sorry my post triggered you. I suppose I should have been more literal. My initial comment was sarcasm.

Musk has made a serious error. The look of the world's richest man butchering government jobs is not great. He has alienated his customer base and sales are imploding. I think you and I are in agreement on this topic, except you have much more personal hate for Musk. I see him as a modern day Howard Hughes - incredibly brilliant, unique, flawed, complex, and in some ways an idiot. The guy is definitely a crappy father and a pathological liar, but the accomplishments at Tesla and SpaceX from an engineering standpoint are absolutely incredible. I don't think the "Nazi salute" meant anything, but that is just my opinion and I am often wrong. As anyone who has studied Musk knows, the guy is on the spectrum and is extremely socially awkward.

As far as public opinion of DOGE - the polls are very mixed. It's largely along political lines. It is WILDLY popular amongst Trump's base (at least for now until the job market fully rolls.)

The reality that most reasonable people (not politicians) can agree we do need to get our fiscal situation in order. Cuts along the lines of DOGE are a part of this. Unfortunately the main areas where this is needed is in rationalizing entitlements while tweaking the tax code.

Natty

Subject Re: Re: Re: Sales imploding globally
Entry 03/10/2025 02:50 PM
Member Rulon Gardner

It kind of is incredible that they are actually cutting government spending which is generally viewed as a sacred cow that is untouchable

No comment on Tesla or Musk, but just marveling that politicians are actually doing it. No comment on Trump as well. No need to start polarizing political rants.

Subject Re: Re: Re: Sales imploding globally
Entry 03/10/2025 03:31 PM
Member ahnuld

I agree Musk is extremely popular amongst Trumps real base (the people who believe the election was stolen in 2020) which I believe is about 30% of the population. Dems hate him. The other 20% probably neutral. But does that 30% Trump base buy EVs?

seems a bit insane that anyone would actually be long Tesla here

Subject Re: Re: Sales imploding globally
Entry 03/11/2025 02:29 PM
Member TOYPAJ182

UCB, if Elon left Tesla, applying your logic, do you really think the stock would be up on that news? My guess is down anywhere between 50%-80%. The auto biz is not the main driver for any bull's thesis, whether it's this writeup we're commenting on, Ron Baron, Cathie, Jeff Skoll, etc. I see what Elon has done with xAI and it's nothing short of remarkable. All that needs to happen with the stock is shift in sentiment away from DOGE / Ukraine comments to v14 / Optimus (weighted more towards FSD) and the stock rebounds from current levels. When was the last time TSLA was "uninvestable"? Typically corresponds with favorable r/r.

Subject Re: Re: Re: Sales imploding globally
Entry 03/11/2025 03:33 PM
Member heffer504

TOYPAJ,

Just checking in (since you dodged the last time I ask) that you have not taken (or booked) any airplane trips since disavowing that obviously inferior mode of transportation?

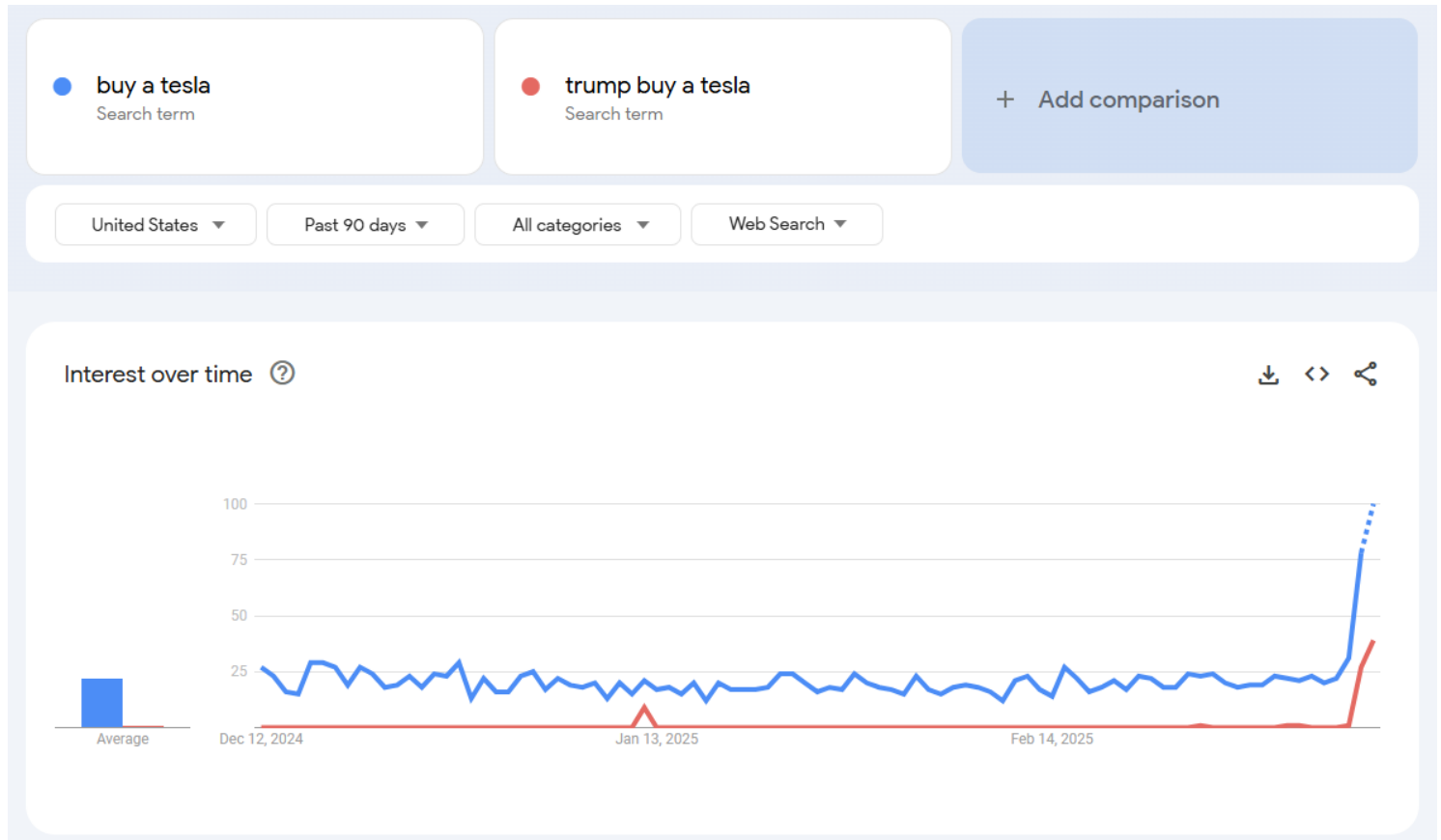
Also, by "nothing short of remarkable", do you mean, "roughly equivalent to what a few dozen Chinese programmers accomplished in a year or so with Deep Seek"? Just trying to calibrate your superlatives...

Regards, Heffer

Subject Re: Re: Re: Re: Sales imploding globally
Entry 03/12/2025 11:51 AM
Member RugPull1995

<https://x.com/sawyermerritt/status/1899664888458580223?s=12&t=J1BtZdaKbJ0WSInmr21YxQ>

Subject	Re: Re: Re: Re: Re: Sales imploding globally
Entry	03/12/2025 12:19 PM
Member	Manchu
I have a stock to sell you	



Subject	Re: Re: Re: Re: Sales imploding globally
Entry	03/12/2025 01:41 PM

Member

T0YPAJ182

Heffer,

Do you think it's a good idea to victory lap / press short with the stock being called uninvestable after being down 50% from its highs? Perhaps EDS? As a short, just know that you will be fighting v14 / Optimus more than you will be fighting unit sales over the next year. Remember, everything is computer.

Also, no air travel until S4 TC.

Subject Re: Re: Re: Re: Re: Sales imploding globally
Entry 03/12/2025 07:26 PM
Member heffer504

Glad to hear you are sticking to your guns on the air travel!

As for the stock, who knows. It seems that if a vast swath of customers will never buy a Tesla (self-driving or not), then the whole debate about whether self-driving is really self-driving (not really a debate-- it's not! but still...) doesn't really matter, since he simply won't sell many cars. Same for the robot which won't work for at least a decade, I think we both know...

Still, I think that Tesla loses \$3.50 this year (and burns \$20B of cash) instead of making \$2, so my inclination is to say that yeah, the stock should go lower. But this is a sui generis situation, and so I think it's really not about unit sales or robots or self-driving, but about bitcoin price and Korean day trader confidence and silly stuff like that.

Subject Tesla Footnotes
Entry 04/22/2025 04:57 PM
Member marwari25

footnotes -- hard to keep track of the changes every quarter!

- (1) Beginning in Q1'25, Capital expenditures is presented inclusive of purchases of solar energy systems and all prior periods have been adjusted.
- (2) As a result of the adoption of the new crypto assets standard, the previously reported quarterly periods in 2024 have been recast.
- (3) Beginning in Q1'25, Adjusted EBITDA (non-GAAP) is presented net of digital assets gains and losses and all prior periods have been adjusted.

Subject Apropos of "What Did I Miss"
Entry 05/13/2025 11:30 PM
Member bowd57

Hi, T0YPAJ182 --

> you will be fighting v14 / Optimus more than you will be fighting unit sales over the next year.

I'm guessing v14 here means FSD Version 14? Or is something completely new? If the former, I'd rather not wade through the sewer that it Tesla discourse to dredge up the details, could you provide a precis? Are there specs, a timeline, that kind of thing?

Thankfully yours,

Bowd

Subject Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves
Entry 09/10/2025 09:46 PM
Member bowd57

Hi, WMC --

Per Grok:

"Based on Elon Musk's All-In Summit remarks, Tesla FSD v14 on HW4 is projected to reach 2-10x human driver safety by late 2025, with a 10x parameter increase for better autonomy. V13 claims focus on fewer interventions, but no specific safety multiple is stated yet."

Yours,

Bowd

Subject Re: Re: FSD version 13 (13.2.1) is now going wide release -- I strongly suggest everyone experiences it for themselves
Entry 09/11/2025 12:20 AM
Member Superflare

Needless to say, I think the release of FSD 14 (supposed to be by end of September in some form, perhaps to early testers) will be a make-or-break moment for Tesla given we have not seen a substantive FSD update released to the public since February (v13.2.8; v13.2.9 was released a few months later, but unclear to me what has changed in that point release). In addition to the timing of 7 months since the last major update, based on the 10x parameter increase & other upcoming features that may be in v14 (i.e., 3x context length which is mentioned in upcoming updates in the v13.2.9 release notes), both of which in theory would require a lot more compute and memory capacity, it is probably fair to assume that v14 will be close to hitting the wall on the HW4 chip's capabilities. If V14 isn't a huge step up, it's possible that even if Tesla's general approach is sound, it will take until HW5/AI5's release bringing a lot more inference horsepower for unsupervised to be feasible.

As a daily user of FSD somewhat versed in the technology, I am optimistic that the upcoming changes (10x parameter count, presumably 3x context length unless they had to abandon that goal due to compute or memory constraints), as well as simply a lot more training done since the last release, should solve a lot of the outstanding issues with v13.2.9. It's quite rare that something actually safety critical comes up with v13.2.9, but most of the issues are related to situations where there is some nuance that FSD doesn't seem to pick up on, and increasing the number of parameters used as inputs by 10x should obviously in theory "fix" those issues as long as the "right" kind of parameters are added & the increase in parameter count doesn't backfire in some way. The one kind of safety critical issue I find occasionally in more rural driving is FSD being overly aggressive turning at stop signs with high speed cross traffic. It is probably largely a behavioral issue that more training or changing model weights should fix, but velocity & distance estimation improvements would also help these situations, and 3x'ing the context length would in theory reduce velocity & distance estimation errors meaningfully (more frames in the context window = better estimation of speed and distance of the object).

But while it's less sexy than 10x parameter count or 3x context length, "simply" more training time & changing weights in the model can clearly have a huge impact on safety. Between v13.2.2 released in December 2024 and v13.2.8 released in February 2025, there was a massive increase in safety in my experience (i.e., following distance and braking behavior was night and day over that period of point releases), but it's unlikely much was done to the model over that period other than changing model weights in response to intervention data.

I'm still skeptical of Elon's commentary on safety relative to a human, but we'll see. This is also very situation dependent. In typical highway driving, I'd guess v13.2.9 could be competitive or safer than a human driver already - it can occasionally do some confusing things that annoy me as a driver and might cause me to take over, but it is never going to depart its lane, it's never not going to stop for traffic ahead because it isn't paying attention, etc. On the other hand, on rural roads with a lot of stop signs, turns into high speed cross traffic, etc, v13.2.9 is clearly below human level of safety.

Subject	FSD/Superflare
Entry	09/22/2025 09:54 PM
Member	bowd57

Hi, Superflare --

Sorry for the slow response, and I think I owe you some responses from an earlier thread. My goal is to have a conversation, not an argument. If it feels like I'm arguing with you, let me know and I'll gladly rethink/rephrase. This has become long, so I'm going to put some bold headers in. There are three sections: Side Remarks, Tracker Analysis, V14.

Side Remarks

Before getting into details, I think you're generally too sanguine about how easy it is to fix serious long-standing problems. Presumably, if they were easy to fix, they would have been. I'm thinking of, for instance, phantom braking, where the system will slam on the brakes at highway speed for no apparent reason. This also happens to other driving assistance systems, I don't know how often, and that it's gotten much better for Tesla lately. But it's a big problem! If I had an ADAS that did that, I'm immediately say, "Please remove all traces of your defective software from my car and give me my money back." Obviously tastes can differ here. But, big problem that bedeviled people for years. Seems like it should be easy to fix? But it wasn't. It gives the impression that there's a knob, and if you dial it one way, you slam into things, but if you dial it the other way, you slam the brakes for nothing, and that there was no setting that would both not slam on the brakes nor slam into things.

A similar story with mapping, which people have been bitching about and blaming things on for years. Should be easy to fix, right? Well, it's not fixed. I don't know why. It wouldn't seem like there would be huge technical challenges. Maybe Musk just doesn't want to, for some reason? Maybe Google Maps is the only better alternative and Musk just doesn't want to deal with them? Beats me. Tesla's very opaque and it's hard for me to judge what's going on. The point of this example is that things could (maybe) wind up un-fixed for non-technical reasons.

I think you also make over-strong statements about FSD's reliability.

> it is never going to depart its lane, it's never not going to stop for traffic ahead

I don't know how often these things happen, but there's video evidence that they do.

> I don't encounter scenarios where it seems like the camera approach failed

I'm not saying this is a safety issue, but there's plenty of video evidence that FSD will get confused by things like skid marks on the road.

Here's a good source for FSD videos:

<https://www.reddit.com/r/TeslaFSD/>

It's remarkably free of the usual hate/fanboy stuff. It's a decent way to keep track of what FSD can still get wrong, however infrequently.

TRACKER ANALYSIS

> I think the release of FSD 14 (supposed to be by end of September in some form, perhaps to early testers) will be a make-or-break moment for Tesla [...] In addition to the timing of 7 months since the last major update,

I agree that what's on the road now is stale. I'm skeptical that v14 will push FSD over the edge, although this is obviously a "Time Will Tell" question. To understand why, we need to (1) get a sense of where we are now, and (2) speculate about how v14 might change things. For #1, let's take a look at the tracker. You've in the past said you placed little faith in the tracker, because it didn't reflect your experience. And I'm sure it doesn't. But it does *encompass* your experience! You're not alone in thinking FSD is the bee's knees and close to perfect. Here are some *states*, not people, that have recorded zero critical disengagements on v13: FL, NC, AR, OR, I'm too lazy to type them all out.

So, here's where we are now:

Miles to CDE	208
# of CDEs	128
Total Miles	26624

CDE Name	Count	Miles/CDE
Traffic Control	77	346
Obstacle	29	918
Ambulance/School Bus	18	1479
Unsafe Manuever	4	6656
RedWheel/Error	38	701

CDE means, "Critical Disengagement". RedWheel/Error is when FSD just suddenly shuts itself, usually due to glare but sometimes due to a crash. For reasons that aren't clear to me, the tracker proprietor doesn't think this counts as critical. I'm going to lump Obstacle & Unsafe into "At Fault Accident" and Traffic Control (running a red light, etc.) and RedWheel/Error into "Liability Risk", and just ignore the rest, although maybe they should go with "Liability Risk". This gives us:

	Count	Miles Between
At Fault Accident	33	807
Liability Risk	115	232

The "Liability Risk" category is -- If an accident ever occurred while FSD was blinded or breaking the law, that would be really bad. And I don't know how easy any of this is to fix. FSD has gotten a *lot* better at not slamming into things, but as I recall, people thought traffic control was a step back with v13. FSD does not handle RedWheel/Error -- at all. There is currently no safe recovery, it goes straight to the driver.

A big question is, "Well, these are all just disengagements. How many of them are legit?" For the Liability Risk category, I'm going with 100%. RedWheel/Error is hard to misinterpret. FSD will gladly ignore red lights and stop signs, and I have no anecdotes about, "It was going to blow the light but then slammed the brakes on at the last second," so I think 100% is close enough there, too.

The tricky part is the At Fault Accident category. No one believes, or should believe, that every CDE prevented a crash. Here's a teensy table showing miles between AFAs for some powers of 10, and another #:

	% Legit?			
Pct	10.00%	1.00%	0.10%	4.00%
Miles/AFA	8,068	80,679	806,788	20,170

0.10% is out because if that was close to where the internal Tesla estimates were, there would be no safety drivers in Austin. We could debate 1% and 10%. I'm going to suggest running with "the other #", 4%, not because I believe it, but rather because it's pulled from an adjacent domain: 4% of trips have a CDE, so why can't 4% of AFAs result in accident? I don't care how dumb that sounds, it's just a number.

Wherever you wind up on, "Where we are," there are two things to keep in mind about, "Where we need to be": (1) I'm framing every "necessary" disengagement as being At Fault, based on video anecdotage. You need to adjust that to get to a miles-between-accident #, they're not the same thing, like, if 4% is the right number, then if every car was using FSD 13 unsupervised, there'd be an accident every 10k miles. (2) For liability purposes -- Mmm. People like to say, "It only needs to be as safe as a human driver, and that's not hard!" The problem is that a lot of accidents are caused by drivers who are breaking the law. Unless you enjoy large punitive damages, you need to be as safe as a law-abiding driver, so raw stats need to be scaled up by at least 50%. So, people like to throw around 500k miles, that needs to be 750k.

V14

I'm really running out of steam. My nicotine and alcohol levels are dangerously low, I've got to wrap this up for my own safety.

> (10x parameter count, presumably 3x context length [...]) "simply" more training time & changing weights

Time Will Tell, but that all sounds like standard scaling stuff to me and think it will result in modest improvements.

From: https://valueinvestorsclub.com/idea/TESLA_INC/3291208721/messages/230222

Some Tesla fans point to scaling laws, generally derived from experience with LLMs, as evidence of the inevitability of Tesla achieving autonomy, but I've never understood why. I'm going to present what I think is the canonical chart of LLM scaling laws:

<post has the chart>

<Snip>

I'll say again, "Somebody's missing something and maybe it's me," I'd love to understand how much data & compute & etc. it would take to move the needle at this point.

So, I don't know ... Yes, more data, more parameters. That could lead to ... 30% increase in performance? Time Will Tell.

Superflare, I'd like to thank you for the SuperCivil conversation.

Yours,

Bowd

Subject	FSD is amazing
Entry	10/15/2025 10:13 AM
Member	kgalt

I've now had FSD 13.2.9 for ~1,000 miles and I probably touched the steering wheel in under 10 of those driven miles.

It's gotten to the point where I can space out and not really focus on the road.

Reportedly this is also the build running a lot of the robotaxis in Austin.

Early Access users who have tried 14.1 and 14.1.1 released over the last week say those versions are even more sentient to the point where the car perceives risks that are impossible for a human to perceive (there are documented examples including a bunny running across the street in total darkness where the driver couldn't see anything but the car paused to avoid roadkill).

Subject	Re: FSD is amazing
Entry	10/15/2025 12:53 PM
Member	Superflare

Jgalt I agree with your commentary on 13.2.9 and continue to be really impressed with it, I'm probably about 20K miles in now on 13.2.8 and 13.2.9, and think the faults it does have seem like things that are fixable by the software rather than problems with the hardware or the general approach of FSD - i.e. they seem like things that will get fixed. I think the majority of the issues I face at this point are navigation related rather than driving related.

I would caution on 14.1, the initial feedback seemed pretty mixed and the jury is still out, but it does appear that within a few days they were able to iron out a lot of the braking issues w/ the 14.1.1 point release. The beginning and end of drive features of v14 are a clear step up and necessary for fully autonomous operation. The actual driving behavior in the middle of the drive seems like it is also a step up in the ways jgalt describes below, i.e. car is reacting faster and taking into account more risks than before, but the growing pains are that because of the # of things the car is now noticing on the road, it is resulting in some harsher braking behavior / excessive caution (whereas 13.2.9 is super smooth and comfortable, but maybe ignores some risks and doesn't have the reaction time that v14.1 seems to have). One of the cool things early access users have pointed out on v14 is its ability to know how to navigate a fast food drive-through (i.e., know to start going once the food has been handed to the driver, without any input). The real world value of the car being able to do that is low or nonexistent, but it points to the v14 model being able to "think" and use context clues in a way that perhaps prior versions of FSD could not. It's unlikely any training was done on drive-through data, so this is being pointed to as a sign the model can "think" outside of its direct training data.

While no data exists yet, my hunch from the videos of 14.1.1 is that the safety of the model - i.e. # of miles it could drive before a safety critical intervention - may be much higher than v13. It appears most of the issues people have had w/ 14.1 and to a lesser extent 14.1.1 are overly cautious behavior that could appear awkward or result in an uncomfortable / jerky ride.

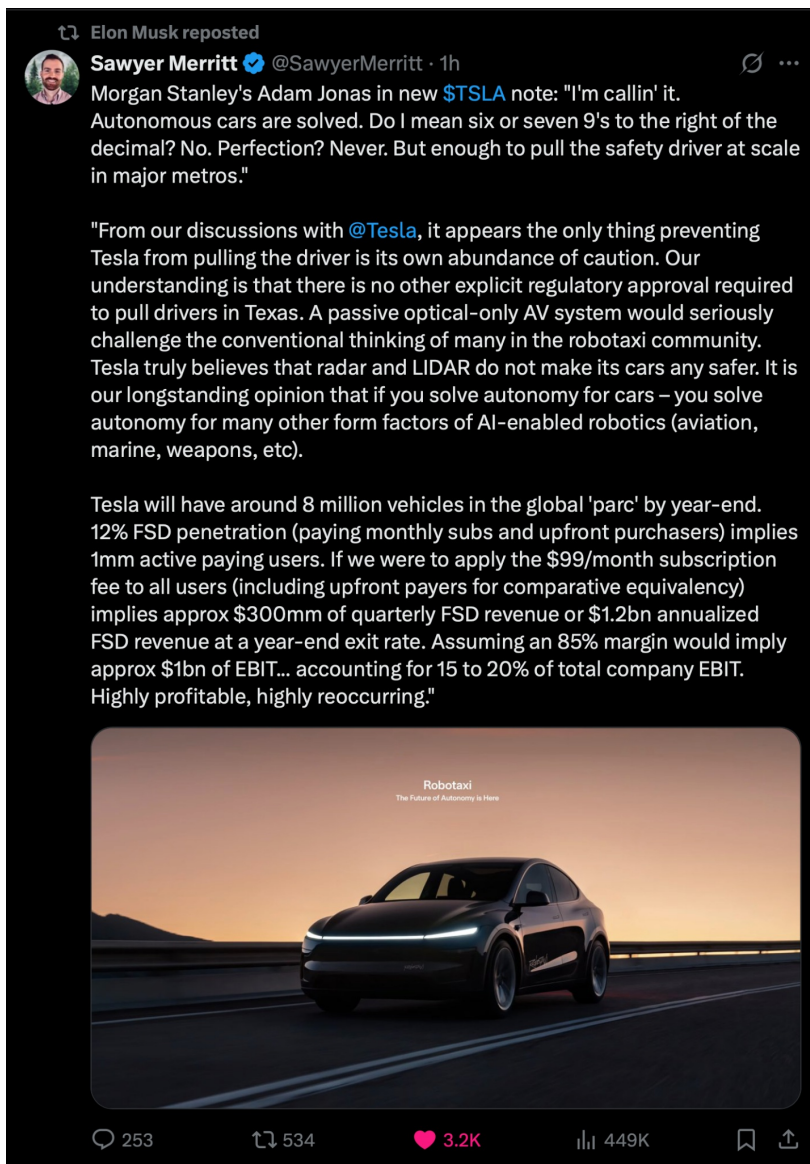
But we'll see soon hopefully when a refined version of v14 reaches the masses like ourselves. Given the wait time between v13 and v14, as I'd said a month ago, v14 better be a huge step up or it is a bad sign.

Subject	Re: Re: FSD is amazing
Entry	10/16/2025 11:13 AM
Member	kgalt

Tesla AI team is cooking. Second point release in a week and now 14.1.2 seems to be going to a wider audience and IS a huge step up.

Subject	FSD is solved
Entry	10/27/2025 12:22 PM
Member	kgalt

also, v14.1.4 started going out last night



<https://x.com/SawyerMerritt/status/1982820970990555351>

Subject
Entry
Member

FSD realtime stats
11/14/2025 04:25 PM
jgalt

<https://www.tesla.com/fsd/safety>

Also, if you haven't yet tried v14.1.4 on MAD MAX mode, I highly recommend scheduling a free test drive: <https://www.tesla.com/drive>

Subject	Re: FSD realtime stats
Entry	11/14/2025 05:12 PM
Member	Superflare

Agreed FSD 14 as of 14.1.4 is really incredible software. While difficult to impossible to quantify this given the data we have (you can look at interventions, but it's so subjective as to what was required vs not, and also so dependent on mix of driving), my guess is FSD 14 is many times safer / capable of driving without getting into an accident than FSD 13, which was already pretty good. There were initially some real rough edges to FSD14 - not actually safety critical in my experience, but harsh braking behavior and some jerkiness that was basically the model being overly sensitive to potential risks like tiny objects on the road that it shouldn't be reacting to - but they are getting smoothed out each build. I have not yet tried FSD 14.1.7, which is only in wide release for Cybertruck, but early reports suggest that is much smoother, and Elon has said FSD 14.2 will return smoothness to more of a FSD13 level.

Either way, my guess is as of 14.1.4 the software may be higher than human level of safety in at least a subset of driving situations, i.e. highway. The reaction time of FSD 14 is just so fast compared to prior versions of the software and probably compared to the average human driver who is often not 100% focused on the road.

Other than the smoothness issues that are getting ironed out, the biggest problem I face w/ FSD is navigation. Not too often, but sometimes the nav is off or FSD makes a navigation mistake and it feels like the driving behavior is now a lot further along than the nav.

If someone has never tried FSD and is nervous about it, I don't know that FSD 14.1.4 is the right build given it can be a bit harsh at times on the brakes, but give it one or two more point releases and I think it'll be time for someone who hasn't tried this software and is skeptical to give it a go.

My gut after experiencing 1,500 miles or so on 14.1.4 is I would put high probability on them getting to above human level of safety with the current (HW4) hardware, in most conditions (snow / heavily salted roads blinding the rear-facing cameras is the type of condition I think they will struggle to get working with the current hardware). And in a year, they're going to have AI5, a much more powerful chip in the car with presumably a better camera suite, hopefully with self-cleaning ability for the non-front facing cameras. Even if they don't get there with HW4, seems highly likely to me (can't prove it of course) that they will get there with AI5 and that is a year or so away.

I struggle to turn the above into an investing takeaway. Perhaps buy Tesla, but at \$1.3 trillion seems like FSD may be priced in at this point and it's more about Optimus? Short auto insurers, perhaps, but they are already heading into a cyclical trough in terms of multiple as people worry about the cycle shifting negatively. Short stuff that depends on collisions i.e. paint for repairs? Just enjoy the technology and invest elsewhere is so far my conclusion.

Subject	Re: FSD realtime stats
Entry	11/15/2025 08:04 AM
Member	heffer504

Musk is literally single-handedly responsible for the murder via starvation of hundreds of thousands, if not millions.

The fact that you're recommending his products makes me nauseous.

Subject	Re:FSD realtime stats
Entry	11/20/2025 09:38 PM
Member	bowd57

Hi, all –

JGalt, Supernova, I appreciate the thoughts.

From my (polar opposite, of course!) POV, I'm not sure what to make of v14. Anecdotal is divided. That rhapsodies are more rhapsodic than usual. But the amount of bitching, whining, and complaining is also much higher.

I'm mostly just surprised. Absent the 10x parameter increase, I would have expected some new features and some incremental improvements.

A problem that I have is that I rely on the tracker to, well, track things. The tracker reports zero critical disengagements for v14. But I'm not sure that's legit because (A) I just wasn't expecting it, and (B) well, this will require a brief diversion into tracker internals to explain. Some time ago Elias, the tracker proprietor, enabled and actually started proselytizing for a device that can automatically report disengagements. That's "report", not "annotate". Uptake has been rapid. "Report" but not "annotate" means that disengagement *causation* is being collected from only 25% of users. Those users also happen to *disengage, for whatever reason*, at about 25% of the rate of everyone else. One of the long standing problems with following/discussing FSD is that it seems to work much better for some people than others. A plausible explanation for what we're seeing in the tracker is that those who are actually reporting causation are drawn from the subset of users for whom FSD actually works.

To be clear, current tracker stats say that v14 is, if not The One, a major step forward. I'm just surprised and somewhat skeptical.

JGalt & Supernova, years of following have also left me skeptical about point releases fixing bugs.

Supernova:> in a year, they're going to have AI5

This is currently slated for mid '27.

J&S, what does valuation look like here to you if the market turns out to be competitive? I'm reminded of Lincoln's Second Inaugural Address:

"Neither party expected for the war the magnitude or the duration which it has already attained. Neither anticipated that the cause of the conflict might cease with or even before the conflict itself should cease. Each looked for an easier triumph and a result less fundamental and astounding. Both read the same Bible and pray to the same God and each invokes His aid against the other [...] The prayers of both could not be answered ~ that of neither has been answered fully. The Almighty has His own purposes."

Say Tesla surprises me by getting this to work in a commercially reasonable time-frame, but doesn't actually sweep the board with 1MM robotaxis waking up overnight. Then what?

Yours,

Bowd

Subject	Re: Re:FSD realtime stats
Entry	11/26/2025 03:22 PM
Member	Superflare

Tesla started releasing FSD 14.2 this past weekend to some users. I have driven about 500 miles now in a variety of environments as well as in medium/heavy rain and dense fog last night. I have yet to have an intervention, of course low sample size. The 14.2 build seems like an incremental improvement in driving behavior over the 14.1 builds & importantly appears to resolve the harsh braking / indecision issues the 14.1 builds introduced.

IMO this is the build where if you have been skeptical of FSD and haven't tried it yet, or haven't tried it in a while, it is time to go schedule a test drive at a Tesla dealer or find a friend with a HW4 Tesla and 14.2 and go for a drive. Is it perfect, no - there are still navigation issues (navigation is now by far the weak link of FSD in my experience), speed control is improved in 14.2 IMO but not going to satisfy everyone, and parking is solid but needs some work. But it is a smooth, seemingly safe build in my experience so far. While you'd need thousands, tens of thousands really, of miles to begin to "prove" the safety of v14.2, I think there is still no substitute for experiencing it first hand.

Bowd: On AI5 timing, you're right that a couple of days after my last post, Elon came out and said AI5 won't be available in mass quantity at least until mid-2027. Previously it was supposed to be by late 2026 / early 2027. Whether this is neutral or bearish for Tesla FSD is hard to say IMO. On one hand, if it turns out AI5 is necessary for an above-human level of safety, then it is obviously bearish to have it delayed. On the other hand, if it turns out HW4 is "enough" - and I am personally gaining confidence in that with the v14 builds - then it may not matter much, and to the extent the delay is at all discretionary, may signal Tesla's confidence in HW4. My gut is HW4 will be "enough" in most conditions but the rear-facing cameras of HW4 vehicles not being able to clean themselves will be an issue when there's snow and particularly salt on the ground, probably requiring human supervision when there's a lot of snow falling / on the ground. The HW4 cameras seem to be adequate in heavy rain, however. I also would bet HW4 cars will never be able to respond to all human instructions, i.e. verbal commands, which could present an issue in some edge cases (although the car does seem to behave well around construction workers / police and pedestrians giving hand gestures, as of v14).

On valuation and competition - I am skeptical of competition for FSD in the near to medium term, but the valuation is really hard to make work even if you are confident in their FSD progress, IMO. That's why I don't own the stock. I think to own Tesla today, you have to either believe they have a near monopoly on self-driving globally (in practice it seems like they'll face meaningful Waymo competition in the U.S. at least for rideshare, and potentially Chinese competitors where they're allowed to operate) or you need to believe in their robotics efforts.

On competition, I think partnering w/ Waymo (or Tesla, but no signs of that yet) is going to be the realistic path forward for other auto OEMs. I don't believe that other auto OEMs have even the camera systems and data systems (how many cars are even regularly connected to wifi to upload data?) in place to begin collecting the kind of fleet data that Tesla has been collecting for 6+ years now and has used to train FSD. I also believe the iterative process of Tesla releasing FSD builds, getting data from interventions, and then using that to fine tune / retrain the model is important. And I don't think Ford or Toyota or other legacy auto OEMs are going to be releasing beta versions of software to the public - that they know have serious faults - to gather data to refine the model. Tesla has a customer base and a corporate risk appetite where they were able to release frankly dangerous software builds to the public and use the data from those builds to get to where they're at now, which at this point I believe is quite safe. I don't think the big, traditional OEMs can or will release software in the state that the early v12 builds of FSD were or certainly the pre-v12 builds were, and I don't think the traditional OEM's average customer would be willing to test beta software either. And then there's Tesla's ability to invest in GPU compute capacity, Tesla's superior engineering/technical employee base, etc that it may be hard for profit-focused, traditional OEMs to replicate.

Subject	Superflare
Entry	12/17/2025 09:14 PM
Member	bowd57

Hi, Superflare --

> IMO this is the build where if you have been skeptical of FSD and haven't tried it yet, or haven't tried it in a while, it is time to go schedule a test drive

This a bit awkward but -- At some level one of the points of wandering talking to people is to know and be known, so I'm just going to put this out there: That statement is completely alien to me. I'm not saying this as a criticism or to start a debate, that's just, ah, just me. I'd like to just leave that there, but in order to fill any potential empty air, I'll chatter for a bit.

One or the reasons is, I just think too out the mainstream for my experiences & personal opinions to be in any way meaningful. All I really care about is babes, beer, board games and basketball, which are all wonderful things, but it leaves me pretty cloistered.

One of my first 10-baggers was Hot Topic, the mall-based retailer of goth fashion to teens. Not only was I never a customer, I don't think I even knew anyone who knew anyone who was a customer. One of the fun things about messing around with stocks is that you can wind up with interesting browsing histories. For a while there, it might have looked like I had a serious Japanese Lolita cosplay fetish, but actually that was just more equity research.

Have we gotten to a point where checkpoints like test drives & trackers are less meaningful? Tesla now claims that things that, in the past, were purely hypothetical albeit just around the corner, they claim that they're actually *doing them* now. I'm wary of switching over to this kind of analysis, because it borders on tea-leaf reading, which I've always been reluctant to do with Tesla because the whole company is so weird.

What does "weird" mean? I don't want to get into it. They won't tell you how many Model Ys they sell.

"Tea-leaf" readings means trying to get from company statements & actions to uhh reality. Tesla's opacity and unique & distinctive communications style have deterred me from going down that road. But as time moves on and the company claims milestones, some issues that have been kicking around for a while have moved to the fore.

The oldest of these is: Why have they not applied for a testing permit in California? If I'd asked that question in 2022, I'd have been told something like, "They don't want to get stuck in red tape while they're still developing the system, they'll do that when the time is ripe." And I would have thought, "Sounds like bullshit but I don't want to argue about it." But now? Wait, is the time not ripe?

The Austin deployment paradoxically resurfaces similar long-standing issues. If I'd asked in 2023, "Why aren't they formally & publicly testing this anywhere?", the answer might have been, "That would be geo-fencing, and they're working on a general solution." Which ... at some level would mean they're sandbagging, right? "Yes, they could be running a fleet of 100 robotaxis somewhere easy, but they just don't want to."

But now they do have a geo-fenced testing program. Where are the robotaxis? The general idea is that each mile/trip/car carries with it some risk of a program-ending mishap, and that's why we didn't have the Austin deployment in '21, '22, '23 or '24. I'm open to the idea that they're sandbagging, like they could stay within their safety budget while doing 10, 100, 10000 trips/week but don't see the point, but how far does one want to take this? I draw the line at I guess 10k, or a fleet of say 100 cars, but anything after, I think we're getting into "No Revenue" territory --

<https://www.youtube.com/watch?v=BzAdXyPYKQo>

Does this make any sense?

Yours,

Bowd

Subject
Entry
Member

Re: Superflare
12/19/2025 01:09 PM
Superflare

Hey bowd - I get the gist of what you're saying, I believe - you don't weight your personal opinion on products much - and agree that's generally a good principle. In this case though, I think there is a lot less subjectivity to how a car drives than most products. There will be preferences on speed and driving behavior to some extent, but one can be fairly objective. I think the most subjective thing here is evaluating the situations where FSD makes a mistake - is it safety critical? Is it easily fixable with future software updates or a permanent limitation of the hardware in the vehicle or software approach?

FSD 14.2.1.25 is the latest release, and is really good in my experience. I have yet to have a safety-related intervention in over 2,000 miles on the 14.2 iterations. I have had interventions due to navigation, speed control, and the inability of the car to read specific signs (i.e., no left turn between 4-6 PM at an intersection). I drove around Manhattan and Brooklyn for a while this week with no issues other than an accelerator tap here or there because the car is a bit too conservative around pedestrians and I don't want to frustrate people behind me. The car handled responded to emergency vehicles and even going through a red light when a traffic cop waved us through (and no lead car to follow, so it appears FSD reacted to the hand signal).

As for why the robotaxi rollout has been slow, two thoughts. First, until FSD14, which only came out two months ago and only became smooth and consumer-ready IMO with the 14.2 release a few weeks ago, I don't think FSD (at least the build for consumers) was good enough for unsupervised operation. The 14.2 iterations are the first where I'd personally be comfortable sitting in the back seat if allowed on at least some rides. So we're going on a month or so of software being available where IMO, the safety and capability of the model might be good enough for unsupervised....if I were Tesla I'd wait another iteration or two to iron out kinks and collect data. Second, there are a lot of real world issues with running a taxi operation beyond having a model capable of safe driving. Creating and running a rideshare app that is reliable, routing the cars and planning for demand, making sure the FSD model pulls over to pick up and let off customers in the right safe areas / where the customer wants, etc. Given how little experience Tesla has so far at doing this, it doesn't surprise me the roll out will be slow.

If Tesla had deployed the software that existed a year ago or two years ago in a robotaxi trial, I guarantee there would have been Tesla at-fault crashes. It wasn't ready yet. Now, it is getting very close IMO, and it might be there from a safety perspective, but I just can't say that for sure with 2K miles driven.

Subject	Re: Re: Superflare
Entry	12/19/2025 02:07 PM
Member	JWF211

The data would suggest that Tesla is nowhere near as safe as a human driver – and this is just within a ringfenced section of Austin that is mapped and under the supervision of both remote and on-premise supervisors.

Tesla reports another Robotaxi crash, even with supervisor as it moves to remove them

<https://electrek.co/2025/12/15/tesla-reports-another-robotaxi-crash-even-with-supervisor/>

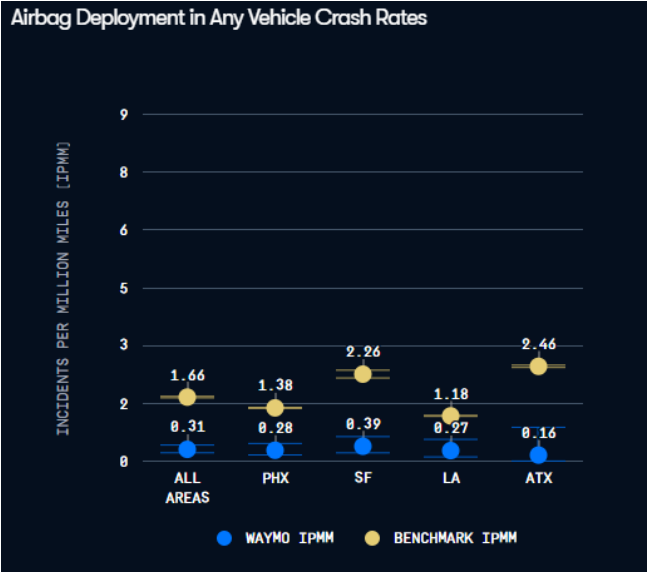
Subject	Re: Superflare
Entry	12/19/2025 02:42 PM
Member	bowd57

Hi, JWF --

> The data would suggest that Tesla is nowhere near as safe as a human driver

I of course agree with the conclusion, but don't think the available data is all that dispositive, if only due to the extremely small amount of data available.

Interestingly, Austin turns out to be a trickier test bed than one might thing. Here's a chart from Waymo:



"ATX" is Austin.

Yours,

Bowd

Subject

Entry

Member

Re: Re: Superflare

12/19/2025 03:26 PM

Superflare

I think the NHTSA data generates some interesting headlines for the Tesla shorts, but it is not a good metric for safety. What counts as an incident here is not what most people would define as a "crash". Many of the details of these 8 incidents are redacted, but for three of them the Tesla was stopped at the time of the "incident." In none of these cases were airbags deployed on either the Tesla or the other vehicle.

I asked ChatGPT Pro to look into the data file for the Waymo data. ChatGPT calculated that in Q3 2025, there was a NHTSA reported incident per 191K miles travelled for Waymo. We "know" from Waymo's robust safety report that depending on the metric you look at, Waymo crashes at 1/5 - 1/10 the rate of human drivers in the same market. So from that, you can conclude that the NHTSA data is not comparable to the metric that Electrek cites, i.e. that there is a human crash per ~500,000 VMT. Using the NHTSA data as a proxy for crashes and taking the 500K figure as the right figure for human drivers, you'd conclude that Waymos are crashing at 2.5x the rate of human drivers, despite robust data from Waymo showing they crash at an apples to apples rate that is 10-20% that of human drivers (~20% total crashes, ~10% the rate of serious crashes).

In addition, based on my personal experience here, the safety of FSD today as of December 2025 is dramatically higher than it was in June 2025, when Robotaxi launched, based on the version of software publicly available. What exactly Robotaxi has been running vs the public builds I can't say.

Subject

Entry

Member

"Time Will Tell"

12/19/2025 04:36 PM

bowd57

Hi, Superflare --

People have been saying "Time will tell" for years. Although the statement itself is neutral, it's usually used as a signal to skeptics that the conversation is over. I'm suggesting that we've gotten to the point where there's really not much else that's meaningful to say.

Again, the bull thesis has two parts:

1: Insuperable cost advantage.

2: Instascale.

The cost advantage comes from two places:

1: Cost of kit needed to deliver the service.

2: Tesla's integrated manufacturing eliminates the need to retrofit the kit.

Which, yeah, those are real if Tesla can get this to work. We could discuss how durable those advantages are, but, in the robotaxi context, if you plug in plausible numbers, you wind up with "competitive cost advantage" not "insuperable cost advantage". \$0.10/mile is a number thrown around by the haters. That's nice but not world-conquering.

That leaves us with instascale. The value of instascale rests on the belief that there are large first-mover advantages, which I wouldn't concede, not because of orneriness, but because of ignorance. Again, for all I know Zoox winds up "winning" because Amazon can bundle it with Prime. NB, this applies to Tesla's competitors, e.g., Waymo, as well. No Tesla bull regards Waymo as having any significant first-mover advantage. And they may well be right.

I'm suggesting that we're really at put-up-or-shut time. I tried at the end of message #166 to switch gears to speculating/reasoning about Tesla internals based on the current state of play in Austin, but if that speculation/reasoning doesn't pretty quick kept swept away by later developments, then ...

I'm going to follow up with a message about attempting to reason/speculate based on the latest consumer release.

Yours,

Bowd

Subject	Re: Re: Re: Superflare
Entry	12/19/2025 05:17 PM
Member	JWF211

Helpful reply! The data, of course, is incomplete. And I don't know how to analogize this data with Waymo's track record. In terms of signal, what matters to me is that 6 out of the 8 incidents resulted in property damage, and one incident involved minor personal injury. The fact that Tesla has already been involved in 6-8 collisions since launching in Austin in late June is not a confidence-inspiring report card. However, the key question remains unanswered: who was at fault?

For AVs to scale, in my opinion, they need to have superhuman safety. Without this condition being met, Tesla's low-price advantage is moot.

Subject	"If You Can See The Progress, Then You're Years Away"
Entry	12/19/2025 05:48 PM
Member	bowd57

Hi, Superflare --

The title quote I think is originally from Brad Templeton. It's apropos of,

> In addition, based on my personal experience here, the safety of FSD today as of December 2025 is dramatically higher than it was in June 2025,

I'm going to present a model. It's tedious and dumb but hopefully not patronizing. Believe it or not, I've been thinking about this for years and this is the best I could come up with.

Say you've got 2MM marbles, 1MM red and 1MM blue. The goal is to remove all the red marbles, leaving only the blue marbles behind.

If the marbles are all spread out on a tray, progress will be pretty linear. Maybe the task will be a bit faster in the beginning, because you might be able to just snatch a whole clump of red marbles, but, whatever: If you've removed 90% of the red marbles, you're 90% done.

Now let's assume that marbles are stored in an urn. You reach in and pluck out a marble. If it's red, you put it aside. If it's blue, you put it in back the urn. That's a very different story. I'm not going to bother to keep the arithmetic straight, but if you're down to only 100k red marbles, you're nothing like 90% done, because you'll need to pluck 10 blue marbles for each red one. When you get down to 10k red marbles, you need to pluck 100 blue marbles, etc.

You could fuck around with the details here, like maybe you the observer don't even know how many marbles there are, you're just watch a glass just getting filled up, etc., but that's the basic idea.

The question is, is developing autonomous automobiles more like "marbles on a tray" or "marbles in an urn"? The industry consensus seems to be, "Marbles in an urn". Elon Musk was doing a simultaneous acknowledgement/hand-wave when he talked about "the march of nines". "Marbles in an urn" resonates with my complete ignorance of the field, because it sounds like what you'd expect if you're using statistical methods.

If industry consensus is right, the fact that you perceived a vast improvement in 6 months makes me ponder marbles & urns and Brad Templeton.

Again, I think we're at a point where all these kinds of speculations will soon be moot. But, because I just can't shut up about Tesla, let me make one final point:

I don't think 14.x is "mature" (whatever that means) enough to be a final candidate. That's because a lot of users perceive it as a large step backwards from 13.x. For whatever reason, I don't care why they think so. That shouldn't happen. "Vodka works."

Yours,

Bowd

Subject	Re: "If You Can See The Progress, Then You're Years Away"
Entry	12/25/2025 09:17 PM
Member	Superflare

Hey bowd - it is an interesting analogy / thought exercise. But I think the Tesla FSD improvement path may be different than the marbles. Key reason comes down to my argument for why Tesla may have a sustainable advantage here - the fact they have been able to deploy beta, potentially dangerous, self-driving software to customers to test and provide disengagement data (I don't think a traditional OEM would be willing to take this reputational risk and I don't think much of a traditional OEM's customer base would be willing to try it either). To use the marble analogy, Tesla is able to deploy its software and get billions of miles travelled on it and get back from the fleet disengagement data, that should tell them exactly where the remaining "red marbles" are. They don't have to try to find the red marbles in some kind of linear or random process that gets more difficult as the ratio of red marbles to blue declines, they are getting back data on exactly where they are (of course, they're also getting back plenty of disengagements that are not safety-related, and have to sort through disengagements to find the ones that are red marbles, but they should be able to use context to figure this out).

My own personal experience w/ FSD has been "exponential" improvement over the past 15 months. In terms of people saying FSD 14 is a step back from FSD 13, I have read people saying that but I think that was (a) on the early builds of 14.1 which were jerky / not as smooth, despite being safer, and/or (b) reflect people being annoyed about non-safety issues w/ FSD 14 (user is no longer able to set a precise max speed, lane change hesitation was annoying on the builds prior to being fixed a few days ago on 14.2.2). If you're in a robotaxi, these things matter less than when it's a driving assist product I think. IMO, the progress from 13->14, as of the 14.2.2 build out now, has been meaningful from a safety perspective. I've personally avoided what I believe would have been a bad accident with someone running a red light that I did not see because they were largely in a blind spot for me, but FSD did see, on FSD 14.1.4, and have had a few other impressive accident avoidance maneuvers on FSD 14 builds (I seem to be encountering a lot of bad human drivers lately!).

BTW, if there's anyone who wants to try out FSD in the NYC area, happy to take anyone for a spin next time I'm around.

Subject	Re: Re: "If You Can See The Progress, Then You're Years Away"
Entry	12/30/2025 12:25 PM
Member	Weighing Machine Capital

@Superflare,

Your analysis is spot on. The barrier to solving autonomy isn't general driving; it's the long tail of edge cases required to reach 99.999999% reliability. Tesla's fleet disengagement data is the defining competitive advantage here — it effectively turns every human intervention into a high-signal label. This represents the largest-scale application of Reinforcement Learning from Human Feedback (RLHF) in history, yet the magnitude of this data advantage is still widely underappreciated.

I am stunned this is not talked about more.

Subject	Re: Re: Re: "If You Can See The Progress, Then You're Years Away"
Entry	12/30/2025 05:01 PM
Member	heffer504

i am stunned this is not talked about more:

<https://www.cgdev.org/blog/update-lives-lost-usaid-cuts>

-heffer

Subject	Re: Re: Re: "If You Can See The Progress, Then You're Years Away"
Entry	12/31/2025 08:37 AM
Member	preyingmantis

a Tesla just drove coast to coast on v14 without a single disengagement, including parking

but this thread will be (m)-debating about how it doesn't work

Gotta love the stock market!

Subject	Re: Re: Re: Re: "If You Can See The Progress, Then You're Years Away"
Entry	12/31/2025 08:41 AM
Member	cuyler1903

FSD is amazing as long as the weather is reasonable. If it starts raining hard, it immediately stops working in my experience.

Subject	Re: "If You Can See The Progress, Then You're Years Away"
Entry	12/31/2025 06:26 PM
Member	bowd57

Hi, all –

WMC:> this data advantage is still widely underappreciated. I am stunned this is not talked about more.

It comes up from time to time.

PreyingMantis:> a Tesla just drove coast to coast on v14 without a single disengagement,

On the other hand, we've got this:

<https://electrek.co/2025/09/21/tesla-influencers-tried-elon-musk-coast-to-coast-self-driving-crashed-before-60-miles/>

I think the real lessons here are about the dangers of anecdote rather than the capabilities of FSD.

Superflare:> <mostly snip>

I think we're getting ever and ever closer to "time will tell".

Superflare:> My own personal experience w/ FSD has been "exponential" improvement over the past 15 months.

It's That Time Of Year, so maybe this is a good time to step back and reassess. Note that I have zero interest in getting into arguments about who was right or wrong about whatever at some point in the past.

Here are some really zoomed out stats from the tracker. I find the tracker useful and relevant for reasons I've explained before and will gladly explain again, although as we'll see it's becoming less useful as FSD progresses:

Version	First Tracker Date	City Miles To Critical DE	Miles To DE	Est. Miles To At Fault Accident:		Cum CDE Mult	Version CDE Mult	CDE CAGR
				At 10% Ratio	At 1% Ratio			
10	10/25/21	74	13	740	7,400	1.00	1	0
11	05/28/23	71	12	710	7,100	0.96	0.96	-2.57%
12	02/21/24	139	30	1,390	13,900	1.88	1.96	31.13%
13	11/30/24	226	31	2,260	22,600	3.05	1.63	43.33%
14	10/07/25	1092	33	10,920	109,200	14.76	4.83	97.56%

Let me know if any of columns need explanation.

Here's some very light annotation:

1: I said something like, "This looks like a project that will never ship" a few months before v12 came out. And you can see why! The product was close to unusable and going nowhere. On getting a look at v12, I immediately retconned the "this team, with this management, and this approach" qualifier in, which I would take to be implicit in any case, because I'm not fond saying dumb things. For context, I regard, "Impossible without LIDAR!" and "Humans drive with vision only!" as being equally dumb.

2: v13 and after only work on newer hardware, which should be accounted for when thinking about the rate of progress of the software.

3: I asterisk the v14 #s for reasons I've explained before and will gladly explain again.

Before getting into interpretation, I'd like to ask Superflare and Weighing Machine Capital and jgalt & anyone else, to provide their own real-time estimates of FSD miles to at-fault accident. Rummage through notes or email or social media posts or whatever. This is not for dunking purposes, but for calibration. Look, I have historical numbers for this stuff. They might be wrong. But they exist, and I can tell you where they came from. If it's not a huge pain in the ass, I'd like to see what other people are using. Superflare reports exponential progress. I'm not arguing with that, I'm just wondering from what base line.

Now for the interpretation and why the tracker is becoming less useful. If you use the v14 numbers with a 1% CDE/at-fault-accident ratio, and the v14 improvement over v13 as the go-forward rate of improvement, then Tesla is there or really close. I'm using 1MM miles/at-fault-accident for Waymo. I'm not sure that you need Waymo's safety levels to make this work. Even if you do, Tesla gets there in a year or so, which is plenty of time for (reasonably) low-risk fleet growth. If you use a 10% ratio and 5x/year, you get there in 3 years. We're getting to the point where questions about tracker accuracy have more salience. I don't want to get into that, other than to observe that everything I can think of points to it being an overestimate of "true" performance, but that the tracker understates the rate of progress because it doesn't account for what I'll call "domain expansion" -- like, Superflare seems to be enjoying using FSD in New York City; if this were 2023, there'd be a YouTube video called, "Superflare takes on the Big Apple!" with 3MM views.

Qualitatively -- and I'm very cynical about Tesla -- I don't think a small scale testing deployment in a single city is a necessarily a stock pump, it seems warranted. Going any further gets us into tea-leaf reading/tracker accuracy, which I don't want to do.

I'd be interested to hear from those who've spent more time thinking about upside scenarios than I have: does Tesla's valuation (ex-Optimus, of course!) require that the autonomous taxi market be uncompetitive?

Yours,

Bowd

Subject
Entry
Member

Of Auto Safety Stats And Trackers
12/31/2025 09:34 PM
bowd57

Hi, all –

I've whined in the past about how challenging I've found it to get/interpret auto safety statistics. This is shit like – I think NHSTA has 2.5MM serious injuries/year, while the National Safety Council is running with 5MM. And it goes on from there. I think you need expertise I don't have to say anything with confidence.

Fortunately, the good folk at Waymo have done a lot of work in this area, written published papers on it, (which as far as I know have not been challenged) and publish easily digestible summaries.

The tracker reports City Miles To DE or CDE. Here, "City" doesn't mean "Urban", it means "Non-Highway." I've always assumed that EV (and thus FSD) usage skews suburban, because EV ownership loses a lot of its appeal if you can't charge at home. I pointed out that tracker stats understate FSD's progress, because they're not adjusted for "domain expansion", that is, using FSD in places where you wouldn't have in the past. I eyeball, but would love to hear different points of view, the impact of this being not all that big, like the surburban/urban split goes from 100/0 to 80/20 or something like that.

The domain effect on tracker reported stats is also hard to guess at. People often use 500k miles/accident. But accident rates swing wildly by locale. Per Waymo, SF has a serious accident every 2MM miles. I think Waymo is using the Abbreviated Injury Scale (guess when I found out what that was?) to define "serious". An example of a serious injury is "open fracture of the humerus". What that means is, the bones from your broken arm are sticking through the skin. That does sound serious! Certainly not a fender-bender. What would an entirely hypothetical 100k miles/at-fault accident FSD rate turn into if converted into Waymo's domain? Who really knows?

I'm posting this just to add some color to things I threw out below. I guess what I'm asking from the FSD advocates is – could you quantify your optimism? Again, this isn't a gotcha question. Miles to at-fault accident might not be your favorite metric. In that case, could you take whatever your favorite metric is, and, with whatever loss of fidelity, translate that into miles to at-fault accident?

Yours,

Bowd

Subject	A different way to think about it?
Entry	01/02/2026 02:56 PM
Member	GCA

A lot of times autonomy is positioned as "will Waymo or Tesla win"? Another way to look at it is, even if Waymo "wins" the robotaxi race, Tesla seems well positioned to dominate autonomous driving capability for consumer-owned cars.

I don't own a Tesla or have a position in the stock, but for interest have been an observer of the autonomous "race". Thus when someone recently asked me for blanket advice when buying a new car, I had to answer honestly that they should consider buying a Tesla solely for the incredible FSD capabilities (judging by anecdotal reports of friends who own... often called "life changing"). Point is, we don't have to wait at all for "time to tell": Tesla has already built an incredible FSD product that people rave about, even if its not robotaxi ready. With that said, judging by FSD progress, it probably will be robotaxi ready soon.

Just some of my probably non-unique observations.

Subject	Re: A different way to think about it?
Entry	01/02/2026 06:35 PM
Member	bowd57

Hi, GCA –

> Tesla seems well positioned to dominate autonomous driving capability for consumer-owned cars.

You have to real careful here. I agree that having an industry leading ADAS package is a differentiator that should help volumes and margins. But –

Selling some more cars isn't all that relevant to a \$1,4B market cap.

You need to be really careful with "autonomous driving capability".

If you mean, "Sleep in the backseat reliability" then, yeah, that would be huge. If you can send your MY out pick up your kids after soccer practice while you watch the Bengals game, then, yes!\$ But that's at least "Robotaxi", and probably a step beyond.

If you mean, "Automation that falls short of sleep in the backseat", then:

A: \$1.4B market cap again.

B: The value proposition is unclear. A lot of people would – and do! – pony up \$8k (or whatever) for what Tesla's got now. Personally, I wouldn't, I just don't find driving that big a chore (not that I drive these days), and there are a lot of people who just can't pony up. Other people are fine with capable highway cruise control/lane-keeping.

C: This a "me" thing. I don't really have a handle on the competitive dynamics in ADAS. (For the acronym-unaware, that's "Advanced Driving Assistance System"; ADS is "Automated Driving System"). I fear I'm becoming long-winded. For instance, China has recently handed out a bunch of Level 3 testing permits to companies that include BYD but not Tesla. I don't want to get hung up on SAE levels, but this is, "Something like what FSD does, but with performance guarantees that Tesla won't currently provide." Industry consensus seems to be, "Taking the driver out is the hard part," so I'm just not sure how durable Tesla's advantage will be.

Yours,

Bowd

Subject	feels like maybe
Entry	01/12/2026 09:05 AM
Member	heffer504

people are going to start realizing that making a huge bet on only cameras is a mistake, and start painting TSLA as an AI loser akin to call centers and Indian outsourcers...

<https://www.bloomberg.com/news/articles/2026-01-12/nvidia-and-mercedes-ai-team-up-makes-other-cars-feel-dumb>

<https://nvidianews.nvidia.com/news/alpamayo-autonomous-vehicle-development>

as self-driving engineering becomes trivial, they may also wonder question why it took Tesla a decade to get to a semi-working model... should be interesting!

Subject	Are We There Yet?
Entry	02/08/2026 06:24 PM
Member	bowd57

Hi, all –

Earlier I suggested that, now that Tesla is actually *doing* it, or at least claiming to, appeals to things like trackers & personal experience of the latest release were less salient because you can instead actually just look at what they're *doing*.

The usual idiots are of course arguing endlessly over every single detail, but it does seem clear to me that there is no evidence for what I call insta-scale. As far as I know, (and I enjoy being corrected! Let me know if I'm off base here!), the thesis is:

1: Insuperable cost advantage, which will be permanent because,

2: All potential competitors are strangled in the crib thanks to insta-scale, due to 1MM robotaxis waking overnight with a single software update.

Nothing I've seen in Austin seems like insta-scale. In Tesla's Q4 presentation, they announce that they're going to expand to 7 additional cities in H1 2026. Maybe that happens, maybe it doesn't, and it's not clear what that means anyway, but "expand to 7 additional cities over the next 6 months" is also not insta-scale.

So again: What's Tesla worth if the market is competitive and slower to evolve than bulls hope? I'm asking because I haven't thought about it much.

Finally, just to add some kind of content, I wandered across this local discussion about Tesla's testing in Austin:

https://www.reddit.com/r/Austin/comments/1qyuwht/tesla_is_running_some_kind_of_secret_operation_in/

What does it all mean? Who knows, but it doesn't sound like insta-scale.

Yours,

Bowd

Subject	Re: Are We There Yet?
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**Entry
Member**

02/09/2026 11:10 PM
Superflare

Hey bowd - while I love to be your sparring partner on Tesla, in this case, I agree with what you wrote. The roll out of unsupervised in Austin has been disappointing. It does appear that an increasing % of robotaxis in Austin are unsupervised (I'm basing this on AIDRIVR, who I follow on X, who has had unsupervised rides for 6 out of his last 7) but the roll out has been slow and there are still growing pains. AIDRIVR posted his first unsupervised ride here and had a bizarre instance of the car swerving to avoid a plastic bag. FSD is prone to sometimes taking overly evasive maneuvers like this for objects in the road, but this seems worse (perhaps indicating higher sensitivity for the unsupervised software than what's on consumer cars right now) than anything I've seen lately. <https://x.com/AIDRIVR/status/2020850297782833394>

In terms of the software in consumer cars, it's been quiet for a while now since 14.2.2 came out in December. Some point releases since then (now on 14.2.2.4) but nothing major in terms of capabilities/performance I've noticed since the 14.2.2 update right before Christmas. 14.3 has been promised as a major update, so we're waiting on that. FSD 14.2.2.4 is really, really good, but still has occasional navigation issues, speed issues in 55 MPH zones, and inability to read general text signs. Have yet to ever have a safety issue on the 14.2 releases though, over many thousands of miles at this point.

In terms of what Tesla is worth, my inclination remains that for one to buy the stock, you need to be bullish on more than FSD and robotaxi. I think you need to be bullish on Optimus to buy the stock at today's price.

**Subject
Entry
Member**

Re: Re: Are We There Yet?
02/10/2026 10:03 AM
natty813

This is possibly the most obscenely overvalued large cap stock in the history of the world. It's so stupid.

They are getting killed in EVs. Anyone without their head completely stuck up their tail can read the WSJ and do any amount of diligence and see that the EXACT same dynamic will exist in humanoid robots, if and when these products actually exist. Have fund competing against 160 government-funded Chinese competitors!

It is also quite clear to me that "Uncle Jimmy" Chanos is likely to be right about self-driving. It's going to work for sure. It's going to be on every single car. It will be like anti-lock brakes or airbags. I don't see excess profits accruing to anyone here. Excess profits accrued to Meta, Google, Microsoft, Apple as these companies DOMINATED their categories and they were monopolies. I don't see those dynamics here at all.

I am not in the shorting business but this is a market dominated by retail investors who understand nothing about competition, supply, demand, etc. I'd wager one day this matters.

**Subject
Entry
Member**

Re: Re: Are We There Yet?
02/10/2026 10:26 AM
acm2025

Can someone explain this to me...my friends in Florida say they can drive to work or 2 hours with only 1 disengagement or whatever. Is that insanely impressive? Is it impressive but still a huge problem because that disengagement would be a wreck or whatever? Does it work in the rain?

**Subject
Entry
Member**

Re: Re: Re: Are We There Yet?
02/10/2026 11:48 AM
LDMR

ACM, correct me if I'm wrong, but I think you're really asking two questions here:

1. Is "FSD" good enough to shuttle people around like Waymo?
2. Does the FSD package provide good enough value to consumers so that Tesla can dramatically outsell competitors.

The objective answer to 1 is no (currently at least). Even if it had a disengagement every 100 hours of drive or 1000 hours of drive it would be way too much. Based on their own numbers they currently crash significantly more than human drivers, even with a human safety monitor (which btw, would be a bad comparison anyway because 40% of crashes in the US are distracted driving or DUI, so if you don't drink and drive / text and drive, your personal risk is even lower than the headline numbers).

The answer for 2 is that you already had your price discovery. At \$100 / month it's a 12% attach rate. And competitors are catching up. Also, the US is a really rich country. \$100 / month is a lot of money even in the richest non-US countries. In Germany it would make an average lease 40% more expensive... (this compares to only 15% more expensive lease in the US)

In any case, we already had price discovery and close to 90% of Tesla owners choose not to subscribe / purchase FSD.

FD 1: I'm biased because I'm obsessed with Tesla's products.

FD 2: I'm biased because I own puts (on natty813's comment #186 thesis).

Subject

Re: Re: Re: Re: Are We There Yet?

Entry 02/10/2026 01:10 PM
Member acm2025

Thanks. Admittedly my questions we not fully formed. I would never buy Tesla for obvious reasons and think it is unshortable for obvious reasons. Mostly just trying to square how much my friends swear by FSD compared with how much the taxi business seems to be struggling.

Subject Re: Re: Re: Re: Re: Are We There Yet?
Entry 02/10/2026 01:14 PM
Member Wrangler

acm, I think you answered your own question when you said "my friends in Florida say they can drive to work or 2 hours with only 1 disengagement or whatever"... You wouldn't use Uber if it killed you "only 1 time or whatever" :)

Subject Re: Re: Re: Re: Re: Re: Are We There Yet?
Entry 02/10/2026 01:22 PM
Member Superflare

Keep in mind there is a difference between a disengagement and something dangerous. I have disengaged plenty of times on the 14.2 branch of software. But I have not yet encountered anything that would have resulted in a crash if I did not disengage. Of course, my sample size while fairly robust is not statistically significant. I've probably driven 10K miles on v14 and maybe 5K on the v14.2 branches. These days, disengagements are almost always related to navigation or failure to change lanes (i.e., FSD is being too conservative and hesitant about changing lanes when it's needed for navigation or to pass someone). On my drive from home to the grocery store, there happens to be a bit of an odd intersection where you need to be in a certain lane. Probably half of the time, FSD doesn't look like it's going to get over fast enough (it realizes late that it's in the wrong lane) so I take over. Wouldn't be dangerous if I didn't, I'd just be going around the block.

Subject Re: Are We There Yet?
Entry 02/10/2026 06:29 PM
Member bowd57

Hi, all –

Me:> Earlier I suggested that, now that Tesla is actually *doing* it, or at least claiming to, appeals to things like trackers & personal experience of the latest release were less salient because you can instead actually just look at what they're *doing*.

Superflare:> I have not yet encountered anything that would have resulted in a crash if I did not disengage.

(Superflare, I'm quoting you purely for rhetorical purposes. I'm going to do more of my usual "no offense meant" stuff after I make my point.)

The whole, at this point multi-year, debate about FSD has been strange to say the least. There are really only two sources of data: the tracker, and personal experience. To those who say, "My personal experience convinces me that Tesla is or will soon be ready for a nation-wide roll out," I'd reply: "Tesla disagrees with you." Obviously, things could change tomorrow. But, for right now, pending further developments, I think the "personal-experience" side needs to move to, "Looks like I was wrong about that, at least for the moment."

No offense meant stuff: Superflare, I appreciate your dialogue throughout. I'm not misrepresenting your views, which are clear, just cherry-picking a quote for punching-bag purposes. I hope I'm also neither grave-dancing nor victory-lapping, things could change overnight. I have zero interest in winning internet debates (note that I don't include making cheap jokes along the way as "winning"): I just want to know if I'm wrong. If I can help a brother out along the way, I'm all for that too.

I'm still skeptical that Tesla will have (or should have! Tesla does things I don't think they should do all the time!) a national robotaxi presence anytime soon, but don't want to get into the trenches about it. I'll just note my esteemed colleague Superflare's opinion that,

> I think you need to be bullish on Optimus

, and reflect on the difference between being right and being early.

Yours,

Bowd

Subject Trenches
Entry 02/10/2026 07:04 PM
Member bowd57

Hi, all –

Just a couple of notes:

Superflare:> It does appear that an increasing % of robotaxis in Austin are unsupervised

This is really trenchy, but – As of a few days ago (note that my obsession level has declined and I'm not tracking as closely as I used to,) all "unsupervised" rides were delivered in a very small area, a single route of 1.5 miles (made that number up), almost as if Robotaxi was running on rails (an in-joke for those who've spent too much time on this). Anyone who knows this can bump their odds of getting "unsupervised" by thoughtful selection of start and finish points.

LDMR:> Even if it had a disengagement every 100 hours of drive or 1000 hours of drive it would be way too much.

As Superflare points out, disengagement <> accident. Note also that i believe the relevant comparable to be *lawful* human driver; using that metric, Waymo's pretty stunning stats (which are flattered by huge outperformance in San Francisco) become more believable.

WRT to Robocar, or whatever Tesla calls its two seater that it's going to make millions of – Fans love to note that 90% of taxi rides are two passengers or less, and the lower cost of a two-seater will ensure that Tesla DOMINATES the space. But things might be a bit more complex. I'm not going to step through the whole thing, but in a competitive market where time to pickup is an important metric, you're turning a 50-50 market into 60-40 if you only run two-seaters. You need to screw around with assumptions to figure out an optimal mix.

I've always found the "Tesla can make millions of robotaxis" an odd argument – how many of these do you think we need in the US? I think you require induced demand to get the # > 1MM.

Yours,

Bowd

Subject	One More Trench
Entry	02/10/2026 07:13 PM
Member	bowd57

Hi, all –

Sorry, forgot about the importance of seasonality and how it could limit insta-scale.

Superflare:> a bizarre instance of the car swerving to avoid a plastic bag.

During the initial rollout of v14, there were widespread complaints about FSD freaking out over blowing leaves. Those complaints have of course vanished as Fall turned into Winter. It's tough to say that what you've got works until you've had it for at least a year.

Yours,

Bowd